

Chapter 3

Consumer theory

3.1 Mathematical aside

3.1.1 Euclidean space and Euclidean distance

A **space** is a list $(Y, t_1, \dots, t_n)$ in which the first component is a set and the other components are “tools” that are available to work in the space. A **metric space** is a set in which one can measure distances between elements, i.e., (Y, d) where Y is a set and $d : X \times X \rightarrow \mathbb{R}$ is a function, called distance, that has the following properties: (1) for each $\{x, y\} \subseteq Y$, $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$; (2) $d(x, y) = d(y, x)$; and (3) for each $z \in Y$, $d(x, y) \leq d(x, z) + d(z, y)$.

Let L be a natural number. The set $\mathbb{R}^L$ is the set of lists of L real numbers. For instance $(1, \sqrt{2}, \pi) \in \mathbb{R}^3$. The generic element in $\mathbb{R}^L$ is denoted $x \equiv (x_l)_{l=1}^L$. The set $\mathbb{R}^L$ is known as the L -dimensional Euclidean space.

Definition 31. For each $\{x, y\} \subseteq \mathbb{R}^L$, the Euclidean distance between x and y is:

$$\|x - y\| \equiv \left(\sum_{l=1}^L |x_l - y_l|^2 \right)^{\frac{1}{2}}$$

In what follows $X \subseteq \mathbb{R}^L$.

Theorem 6. Let $d_E : X \times X$ be the function $(x, y) \in X \times X \mapsto d_E(x, y) \equiv \|x - y\|$. Then, the space (X, d_E) is a metric space, i.e., d_E is a distance.

Definition 32. Let $x \in X$ and $\varepsilon \in \mathbb{R}_{++}$. The open ball centered at x with radius ε in X is the set: $B_\varepsilon^X(x) \equiv \{y \in X : d_E(x, y) < \varepsilon\}$.

Definition 33. A set $A \subseteq X$ is **open in X** if for each $x \in A$ there is $\varepsilon \in \mathbb{R}_{++}$ such that $B_\varepsilon^X(x) \subseteq A$.

Definition 34. A set $A \subseteq X$ is **closed in X** if its complement in X is open, i.e., if $X \setminus A$ is open in X .

Remark 3. A closed set is not necessarily a set that is not open. There are sets that are neither closed, nor open. For instance, $(0, 1]$ is not open or closed in $\mathbb{R}$. There are sets that are closed and open simultaneously. Could you give an example in $\mathbb{R}^L$?

Definition 35. A sequence $\{x^t\}_{t \in \mathbb{N}}$ in X is a function from $\mathbb{N}$ into X . The value of the sequence at $t \in \mathbb{N}$ is $x^t \equiv (x_1^t, \dots, x_L^t) \in X$.

Definition 36. A sequence $\{x^t\}_{t \in \mathbb{N}}$ in X converges to $x \in X$ if for each $\varepsilon \in \mathbb{R}_{++}$ there exists $T(\varepsilon) \in \mathbb{N}$ such that for each $t \in \mathbb{N}$, $t \geq T(\varepsilon)$, $d(x^t, x) < \varepsilon$. If $\{x^t\}_{t \in \mathbb{N}}$ converges to x we write $x^t \xrightarrow[t \rightarrow \infty]{} x$ and define $\lim_{t \rightarrow \infty} x^t \equiv x$.

Lemma 1. A set $G \subseteq X$ is closed in X if and only if for each convergent sequence $x^t \xrightarrow[t \rightarrow \infty]{} x$, such that for Each $t \in \mathbb{N}$, $x^t \in G$, we have that $x \in G$.

Definition 37. A set $H \subseteq X \times X$ is closed in $X \times X$ if for any two convergent sequences in X , $x^t \xrightarrow[t \rightarrow \infty]{} x$ and $y^t \xrightarrow[t \rightarrow \infty]{} y$ such that for each $t \in \mathbb{N}$, $(x^t, y^t) \in H$, we have that $(x, y) \in H$.

3.1.2 Inner product and half spaces

Definition 38. Let $x \equiv (x_l)_{l=1}^L \in \mathbb{R}^L$ and $y \equiv (y_l)_{l=1}^L \in \mathbb{R}^L$. The inner product between x and y is defined by:

$$x \cdot y \equiv x_1 y_1 + \dots + x_L y_L.$$

Definition 39. Let $p \equiv (p_l)_{l=1}^L \in \mathbb{R}^L$ and $w \in \mathbb{R}$. The hyperplane associated with p and w is the set:

$$h(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x = w\}.$$

Each hyperplane determines two closed half spaces and two open half spaces:

Definition 40. Let $p \equiv (p_l)_{l=1}^L \in \mathbb{R}^L$ and $w \in \mathbb{R}$.

- The positive closed half-space associated with p and w is the set:

$$h^{\geq}(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x \geq w\}.$$

- The negative closed half-space associated with p and w is the set:

$$h^{\leq}(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x \leq w\}.$$

- The positive open half-space associated with p and w is the set:

$$h^{>}(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x > w\}.$$

- The negative open half-space associated with p and w is the set:

$$h^{<}(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x < w\}.$$

Geometrically, p is the vector normal to $h(p, w)$. Let $x \in h(p, w)$. If $x' \in h^{>}(p, w)$ then $p \cdot (x' - x) > 0$. If $x' \in h(p, w)$, then $p \cdot (x' - x) = 0$. If $x' \in h^{<}(p, w)$, then $p \cdot (x' - x) < 0$.

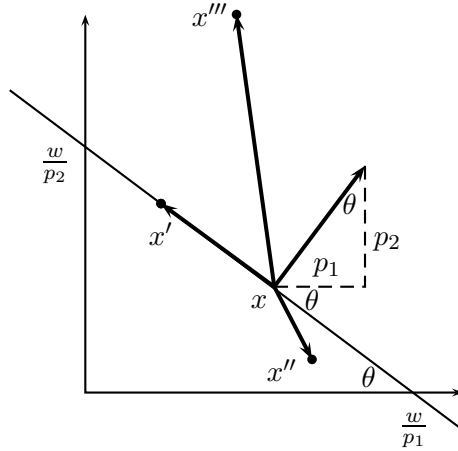


Figure 3.1: half spaces.

3.2 Consumer theory I: Preferences

3.2.1 Alternatives space

- We study the behavior of a single agent when alternatives are consumption bundles of $L \in \mathbb{N}$ commodities.
- The set of alternatives is $X \subseteq \mathbb{R}_+^L$.
- The generic consumption bundle is a vector $x \equiv (x_1, \dots, x_L) \in \mathbb{R}_+^L$; for each $l = 1, \dots, L$, x_l is the amount of commodity l consumed at x . Sometimes it will be useful to write a consumption vector x in matrix notation as $x \equiv [x_1, \dots, x_L]^T$.
- For simplicity we will usually assume that $X \equiv \mathbb{R}_+^L$.

We will study: (1) consumer preferences, (2) utility representation of consumer preferences, (3) consumer choice, and (4) revealed preference theory in the consumer's world.

3.2.2 Consumer preferences

First we study the representation of a consumer by a preference relation $\succsim$ on X .

Definition 41. Let $\succsim$ be a binary relation on $X \subseteq \mathbb{R}^L$ and $x \in X$. The **upper contour set of $\succsim$ at x** is $U(\succsim, x) \equiv \{y \in X : y \succsim x\}$. The **indifference set of $\succsim$ at x** is $I(\succsim, x) \equiv \{y \in X : y \sim x\}$. The **lower contour set of $\succsim$ at x** is $L(\succsim, x) \equiv \{y \in X : x \succsim y\}$. The **strict upper and lower contour sets of $\succsim$ at x** are $U(\succ, x)$ and $L(\succ, x)$, respectively.

The following are further restrictions that are usually assumed on consumer preferences.

Desirability assumptions

Definition 42. We use the following notation for vector inequalities: for each $\{x, y\} \subseteq \mathbb{R}^L$, $\mathbf{x} \geq \mathbf{y}$ if for each $i = 1, \dots, L$, $x_i \geq y_i$; $\mathbf{x} \gneq \mathbf{y}$ if $\mathbf{x} \geq \mathbf{y}$ and $\mathbf{x} \neq \mathbf{y}$; $\mathbf{x} \gg \mathbf{y}$ if for each $i = 1, \dots, L$, $x_i > y_i$.

Definition 43. A binary relation $\succsim$ on $X \subseteq \mathbb{R}^L$ is:

- **strictly monotone** if for each $\{x, y\} \subseteq \mathbb{R}^L$ such that $\mathbf{x} \gneq \mathbf{y}$, we have that $\mathbf{x} \succ \mathbf{y}$ (strict monotonicity is also referred to in the literature as “strong monotonicity.”)

- **monotone** if for each $\{x, y\} \subseteq \mathbb{R}^L$ such that $x \gg y$, we have that $x \succ y$.
- **weakly monotone** if for each $\{x, y\} \subseteq \mathbb{R}^L$ such that $x \geq y$, we have that $x \succeq y$.
- **locally non-satiated** if for each $x \in X$ and each $\varepsilon \in \mathbb{R}_{++}$ there is $y \in B_\varepsilon^X(x)$ such that $y \succ x$.

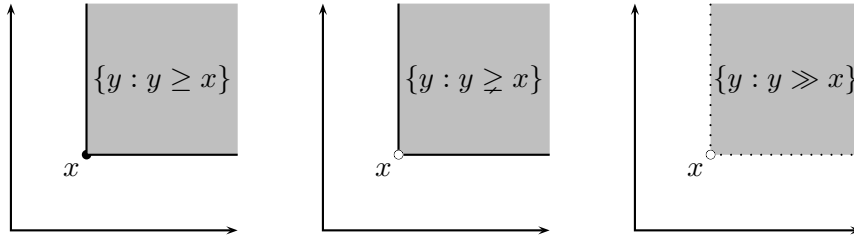


Figure 3.2: Definitions of $\geq$, $\not\geq$, and $\gg$: the three domination notions induce different desirability notions for preferences.

Remark 4. Weak monotonicity does not imply local non-satiation. For instance let $\succeq$ be a preference on $X \equiv \mathbb{R}_+^L$ defined by: for each $\{x, y\} \subseteq X$, $x \sim y$. Clearly, $\succeq$ is weakly monotone, but it is not locally non-satiated.

Remark 5. If $\succeq$ is transitive, weakly monotone, and locally non-satiated, then $\succeq$ is monotone. We prove this. Let $\{x, y\} \in \mathbb{R}_+^L$ be such that $y \gg x$. We prove that $y \succ x$. Suppose by means of contradiction that $\neg y \succ x$. Since $y \gg x$ and $\succeq$ is weakly monotone, then $y \succeq x$. Since $\neg y \succ x$, then $x \succeq y$. Thus, $x \sim y$. We claim that for each $z \in \mathbb{R}_+^L$ such that $x \leq z \leq y$, $z \sim x$. To prove this, observe that since $\succeq$ is weakly monotone, then $y \succeq z \succeq x$. Since $x \succeq y$, then by transitivity of $\succeq$, $x \sim z$.

Let $z^* \equiv \frac{x+y}{2}$. Since $y \gg x$, then $y \gg z^* \gg x$. Let $\varepsilon \equiv \min_{i=1, \dots, L} \frac{|y_i - x_i|}{2}$. We claim that $B_\varepsilon^X(z^*) \subseteq \{z \in \mathbb{R}_+^L : x \leq z \leq y\}$. See figure.

To prove this, let $z \in B_\varepsilon^X(z^*)$. We claim that $x \leq z$. Suppose by means of contradiction that $x \not\leq z$. Thus, there is $k \in \{1, \dots, L\}$ such that $z_k < x_k$. Since $z_k^* > x_k$, then $|z_k^* - z_k| \geq |z_k^* - x_k| \geq \varepsilon$. Now, $\|z^* - z\| =$

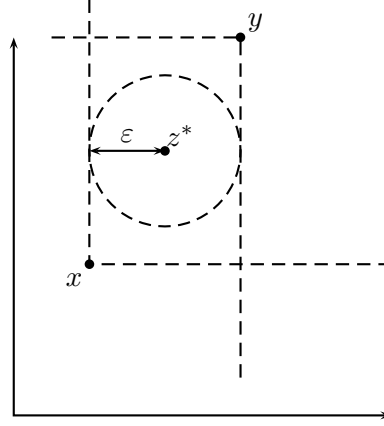


Figure 3.3: Illustrating Remark 5.

$\left(\sum_{l=1}^L (z_l^* - z_l)^2\right)^{\frac{1}{2}} \geq |z_k^* - z_k|$. Thus, $\|z^* - z\| \geq \varepsilon$ and $z \notin B_\varepsilon^X(z^*)$. This is a contradiction. Thus, $x \leq z$. A symmetric argument shows that $z \leq y$.

We conclude that there is $z^* \in \mathbb{R}_+^L$ and $\varepsilon \in \mathbb{R}_{++}$ such that for each $z \in B_\varepsilon^X(z^*)$, $x \leq z \leq y$ and consequently, $z \sim x \sim z^*$; thus, there is no $z \in B_\varepsilon^X(z^*)$ such that $z \succ z^*$. This contradicts local non-satiation of $\succsim$. $\square$

Convexity assumptions

Definition 44. A set $Y \subseteq \mathbb{R}^L$ is **convex** if for each $\{x, y\} \subseteq Y$ and each $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)y \in Y$. Here αx is the vector $(\alpha x_1, \dots, \alpha x_L)$.

Assume that X is a convex set.

Definition 45. A binary relation $\succsim$ on X is **convex** if for each $x \in X$, $U(\succsim, x)$ is convex.

Definition 46. A binary relation $\succsim$ on X is **strictly convex** if for each $x \in X$, each $\{y, z\} \subseteq U(\succsim, x)$, and each $\alpha \in (0, 1)$, $\alpha y + (1 - \alpha)z \succ x$.

Continuity assumptions

Definition 47. A binary relation $\succsim$ on X is **continuous** if it is a closed subset of $X \times X$, i.e., for any two convergent sequences $x^t \xrightarrow[t \rightarrow \infty]{} x$ and $y^t \xrightarrow[t \rightarrow \infty]{} y$ such that for each $t \in \mathbb{N}$, $x^t \succsim y^t$, we have that $x \succsim y$.

Definition 48. A binary relation $\succsim$ is **continuous*** if for each $x \in X$, $U(\succsim, x)$ and $L(\succsim, x)$ are closed in X .

Theorem 7. A preference relation is continuous if and only if it is continuous*.

Proof. We prove both directions independently.

1. ($\Rightarrow$) Assume that $\succsim$ is continuous. Prove that it is continuous*.
2. ($\Leftarrow$) Assume that $\succsim$ is continuous*. We prove that it is continuous, i.e., for any two convergent sequences $x^t \xrightarrow[t \rightarrow \infty]{} x$ and $y^t \xrightarrow[t \rightarrow \infty]{} y$ such that for each $t \in \mathbb{N}$, $x^t \succsim y^t$, we have that $x \succsim y$. Suppose by means of contradiction that $y \succ x$. We prove that there is an open set containing y , $V(y)$, and an open set containing x , $V(x)$ such that for each $y' \in V(y)$ and each $x' \in V(x)$, $y' \succ x'$. There are two cases.
 - (a) There is $z \in X$ such that $y \succ z \succ x$. Prove the claim is true in this case. (You have to: identify candidates for $V(y)$ and $V(x)$; prove these sets are open, $y \in V(y)$, and $x \in V(x)$; prove that they satisfy the desired property.)
 - (b) There is no $z \in X$ such that $y \succ z \succ x$. Prove the claim is true in this case. Hint: let $V(y) \equiv \{y' \in X : y' \succ x\}$ and $V(x) \equiv \{x' \in X : y \succ x'\}$. (You have to: prove these sets are open, $y \in V(y)$, and $x \in V(x)$; prove that they satisfy the desired property.)

Finally conclude that there is a contradiction: what is wrong with the fact that there is an open set containing y , $V(y)$, and an open set containing x , $V(x)$ such that for each $y' \in V(y)$ and each $x' \in V(x)$, $y' \succ x'$? (This statement is not trivially contradictory. Your job consists on finding some consequences of it that are.)

□

Corollary 1. A preference relation $\succsim$ on X is continuous if and only if for each $x \in X$, $U(\succ, x)$ and $L(\succ, x)$ are open in X .

Proposition 7. If $\succsim$ is represented by a continuous utility function, then $\succsim$ is continuous. (A function $u : X \rightarrow \mathbb{R}$ is continuous if for each convergent sequence in X , $x^t \xrightarrow[t \rightarrow \infty]{} x$, we have that $u(x^t) \xrightarrow[t \rightarrow \infty]{} u(x)$.)

Proof. Problem 5, Section 3.2.3.

□

Parametric and simplicity assumptions

In applied work it is useful to restrict attention to families of preferences that can be generated by one of its indifference sets.

Definition 49. A preference relation $\succsim$ is **homothetic** if its indifference sets are related by proportional expansion along rays, i.e., for each pair $\{x, y\} \subseteq X$ such that $x \sim y$ and each $\alpha \in \mathbb{R}_+$, $\alpha x \sim \alpha y$.

Definition 50 (Mas-Colell et al. (1995)). The preference relation $\succsim$ on $X \equiv \mathbb{R} \times \mathbb{R}_+^{L-1}$ is **quasilinear with respect to commodity 1** if:

- If $(x_1, \dots, x_L) \sim (y_1, \dots, y_L)$ then for each $\alpha \in \mathbb{R}$, $(x_1 + \alpha, \dots, x_L) \sim (y_1 + \alpha, \dots, y_L)$.
- For each $(x_1, \dots, x_L) \in X$ and each $\alpha \in \mathbb{R}_{++}$, $(x_1 + \alpha, \dots, x_L) \succ (x_1, \dots, x_L)$.

A preference relation $\succsim$ on $X \equiv \mathbb{R} \times \mathbb{R}_+^{L-1}$ is **quasilinear with respect to commodity l** if the above conditions hold with respect to commodity l .

Definition 51. A preference relation $\succsim$ on $X \subseteq \mathbb{R}_+^2$ is a **Cobb-Douglas** preference if it is represented by a utility function u of the form, $x \equiv (x_1, x_2) \in X \mapsto u(x) \equiv x_1^\alpha x_2^{1-\alpha}$ for some $\alpha \in (0, 1)$.

Definition 52. A preference relation $\succsim$ on $X \subseteq \mathbb{R}_+^2$ is a **CES** (Constant Elasticity of Substitution) preference if it is represented by a utility function u of the form $x \equiv (x_1, x_2) \in X \mapsto u(x) \equiv (\alpha_1 x_1^\rho + \alpha_2 x_2^\rho)^{\frac{1}{\rho}}$ for some $\{\alpha_1, \alpha_2\} \in \mathbb{R}_{++}$ and $\rho \in \mathbb{R} \setminus \{0\}$.

Definition 53. A preference relation $\succsim$ on $X \subseteq \mathbb{R}^L$ is **linear** if it is represented by a utility function u of the form $x \in \mathbb{R}^L \mapsto u(x) \equiv q \cdot x$ for some $q \in \mathbb{R}^L$.

Definition 54. A preference relation $\succsim$ on $X \subseteq \mathbb{R}_+^L$ is the **symmetric Leontief** preference if it is represented by the utility function u defined by $x \equiv (x_l)_{l=1}^L \in \mathbb{R}_+^L \mapsto u(x) \equiv \min_{l=1, \dots, L} x_l$.

Definition 55. Let $f \equiv (f_l)_{l=1}^L$ be a list of strictly increasing functions $f_l : \mathbb{R}_+ \rightarrow \mathbb{R}$. A preference relation $\succsim$ on $X \subseteq \mathbb{R}^L$ is the **generalized Leontief preference associated with f** if it is represented by the utility function u defined by $x \in \mathbb{R}^L \in X \mapsto u(x) \equiv \min_{l=1, \dots, L} f_l(x_l)$.

3.2.3 Problems

1. Assume $X \equiv \mathbb{R}_+^L$. Prove that: (a) Strict monotonicity implies monotonicity; (b) Monotonicity implies local non-satiation; (c) The converse statements are not true.
2. Recall that a preference $\succsim$ on $\mathbb{R}_+^L$ is convex if for each $x \in \mathbb{R}_+^L$, $U(\succsim, x)$ is convex. Assume that $\succsim$ is rational. Prove that $\succsim$ is convex if and only if for each pair $\{x, y\} \subseteq \mathbb{R}_+^L$ such that $x \succsim y$ and each $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha)y \succsim y$.
3. Complete the omitted details in the proof of Theorem 7.
4. Prove that continuity and continuity* are not necessarily equivalent for binary relations that are complete but not transitive, that is, construct a complete and continuous* binary relation that is not continuous. Hint: Let $X \equiv [0, 1]$. Construct a binary relation on X that is incomplete, continuous* but not continuous (this is relatively simple). If you are lucky, you may be able to complete the preference and retain continuity* without gaining continuity. The constructed preference cannot be transitive (You just proved it in Problem 3). A drawing here does the job. Accompany it with a proof that the binary relation is continuous* but not continuous.
5. Prove Proposition 7.
6. Construct a quasilinear preference that is not continuous.
7. Notation: for each pair $\{a, b\} \subseteq \mathbb{R}_+^2$ such that $a \leq b$, let $\text{rec}\{a, b\}$ be the rectangle that has corners a and b , that is, $\{x \in \mathbb{R}_+^2 : a \leq x \leq b\}$. Let $i \in N$ and $\succsim_i$ be a preference relation on $\mathbb{R}_+^2$. Define the sacrifice of agent i at a bundle $z \in \text{rec}\{0, \Omega\}$, denoted $S(\succsim_i, z)$, as the size of the set of bundles that the agent weakly prefers to z and could receive at some feasible allocation, i.e.,

$$S(\succsim_i, z) \equiv \text{size}(\{x \in \mathbb{R}_+^2 : x \leq \Omega, x \succsim_i z\}).$$

Formally, the size of a set here is the Lebesgue measure of the set. Your intuition of the size of the set is sufficient for solving the following questions. You should know for instance that if $A \subseteq B$, $C \subseteq B$, $A \cap C = \emptyset$, and $\text{size}(C) > 0$, then $\text{size}(B) > \text{size}(A)$. Figure 3.4 illustrates the definition. The sacrifice of $\succsim_i$ at z is the size of the shaded area.

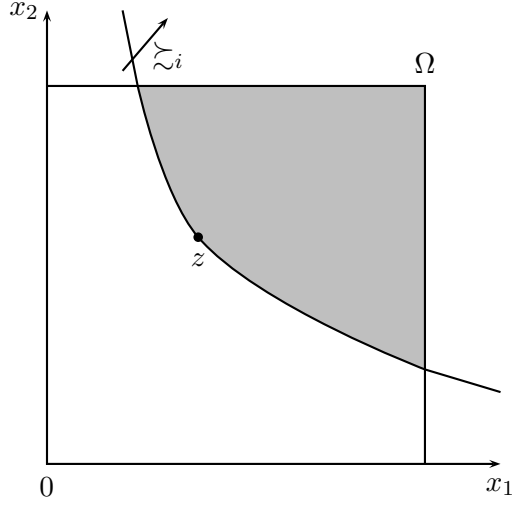


Figure 3.4: Sacrifice of $\tilde{z}_i$ at z .

Consider the following function:

$$\begin{aligned} u_i : \quad \mathbb{R}_+^2 &\rightarrow \mathbb{R} \\ x \equiv (x_1, x_2) &\mapsto u_i(x) \equiv -S(\tilde{z}_i, x). \end{aligned}$$

Prove that if $\tilde{z}_i$ is rational, monotone and continuous, then u_i represents $\tilde{z}_i$ in $\text{rec}\{0, \Omega\}$. That is, for each pair $\{x, y\} \subseteq \text{rec}\{0, \Omega\}$, $x \tilde{z}_i y$ if and only if $u_i(x) \geq u_i(y)$.

8. Consider the preferences on $[1, 2] \subseteq \mathbb{R}$ that are represented by the functions in Figure 3.5.
 - (a) Is the preference in Figure 3.5 (a) continuous? Formally support your answer.
 - (b) Is the preference in Figure 3.5 (b) continuous? Formally support your answer.
9. Suppose that the set of alternatives is $\mathbb{R}_+$. A preference R on $\mathbb{R}_+$ is **single peaked** if there is $x^* \in \mathbb{R}_+$ such that:¹

¹An alternative and common generic notation for a preference relation is R . The asymmetric part of R is usually denoted P . The symmetric part of R is usually denoted I .

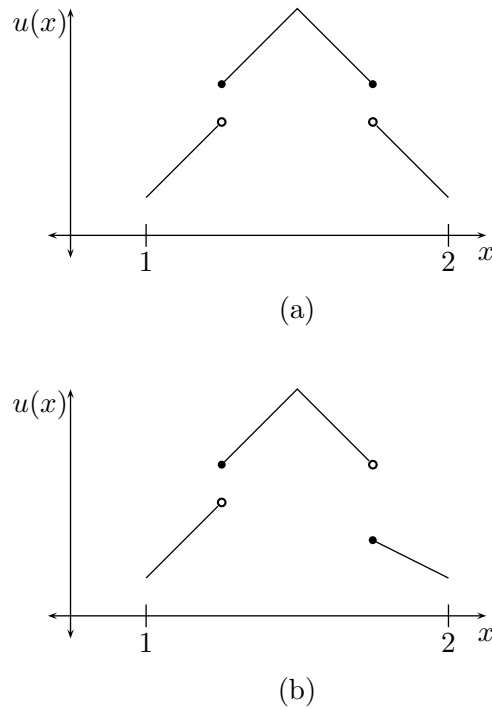


Figure 3.5: Utility functions in Problem 8

- For each pair $\{x, y\} \subseteq \mathbb{R}_+$ such that $x < y \leq x^*$, $y P x$, and
 - For each pair $\{x, y\} \subseteq \mathbb{R}_+$ such that $x^* \leq y < x$, $y P x$.
- (a) Show that if R is continuous then it can be represented by a function u such that for each $x \leq x^*$, $u(x) = x$.
- (b) Show that if R is not continuous then it may not be represented by a function u such that for each $x \leq x^*$, $u(x) = x$. (Here you have to construct a preference that is single peaked but does not admit a representation u such that for each $x \leq x^*$, $u(x) = x$.)
- (c) Describe a situation in which it is reasonable to assume that the set of alternatives is $\mathbb{R}_+$ and preferences are single peaked.
10. Let $\succsim$ be a continuous preference on $\mathbb{R}_+^L$. Let $\{x, y\} \subseteq \mathbb{R}_+^L$ and $p \in \mathbb{R}_{++}^L$. Show that if $p \cdot x \geq p \cdot y$, $y \succ x$, and $x \succsim 0$, then there is $y' \in \mathbb{R}_+^L$ such that $p \cdot y' < p \cdot x$ and $y' \succ x$.

11. Let $X \equiv \mathbb{R}_+^L$. Let $\succsim$ be a monotone and continuous rational preference relation on $\mathbb{R}_+^L$. Suppose that $\succsim$ is such that there is at least one strict preference statement, formally, that there are $\{x, y\} \subseteq \mathbb{R}_+^L$ such that $x \succ y$.
 - (a) Is there a discontinuous utility representation of $\succsim$? Justify your answer.
 - (b) Would your answer change if we do not assume that there is at least one strict preference statement? Justify your answer.
12. Let $\succsim$ be a monotone and continuous preference relation on $\mathbb{R}_+^L$. Prove that if $x \geq y$ then $x \succsim y$.
13. Let $\succsim$ be a continuous preference relation on $\mathbb{R}_+^L$. Let $\{x, y, z\} \subseteq \mathbb{R}_+^L$ be such that $x \succsim y \succ z$. Prove that there is $\alpha \in [0, 1]$ such that $\alpha x + (1 - \alpha)z \sim y$. Prove that under the further conditions that $\succsim$ is strictly convex and $x \succ y \succ z$, then α is unique. Hint: let $\varphi : [0, 1] \rightarrow \mathbb{R}$ be such that $\varphi(0) \leq 0 \leq \varphi(1)$. Then by the Intermediate Value Theorem, there is $\alpha \in [0, 1]$ such that $\varphi(\alpha) = 0$. (Illustrate with a figure).
14. Let $\succsim$ be a monotone and continuous preference relation on $\mathbb{R}_+^L$. Let $p \in \mathbb{R}_{++}^L$, $w \in \mathbb{R}_+$, and $\{x, y\} \subseteq \mathbb{R}_+^L$ be such that $0 < p \cdot x \equiv p_1 x_1 + \cdots + p_L x_L \leq p \cdot y$ and $x \succ y$. Prove that there is $x' \in \mathbb{R}^L$ such that $p \cdot x' < p \cdot x$ and $x' \succ y$. (Illustrate with a figure).
15. Let $\succsim$ be a monotone and continuous preference relation on $\mathbb{R}_+^L$. Let $p \in \mathbb{R}_{++}^L$, $w \in \mathbb{R}_+$, and $\{x, y\} \subseteq \mathbb{R}_+^L$ be such that $p \cdot x < p \cdot y$ and $x \succsim y$. Prove that there is $x' \in \mathbb{R}^L$ such that $p \cdot x' < p \cdot y$ and $x' \succ y$. (Illustrate with a figure).

3.3 Consumer Theory II: Utility

In Section 2.2, we proved that if X is finite then each preference relation on X admits a utility representation. In the consumer's world, X is not finite (not even countable). Thus, this result falls short in this context. Are there preferences in $\mathbb{R}_+^L$ that do not admit a representation? The following example settles this issue.

Example 10. Let $X \equiv \mathbb{R}^2$ and define $\succsim_{lex}$ as follows: for each $x \equiv (x_1, x_2) \in \mathbb{R}^2$ and each $y \equiv (y_1, y_2) \in \mathbb{R}^2$, $x \succsim_{lex} y$ if and only if either $x_1 > y_1$ or $[x_1 = y_1 \text{ and } x_2 \geq y_2]$. One can prove that: $\succsim_{lex}$ is a preference relation that does not admit a utility representation, c.f., [Rubinstein \(2006\)](#).

3.3.1 Existence of continuous representation

There are very general conditions under which we can guarantee a continuous preference admits a continuous representation. Loosely speaking they are topological properties of X . We will prove a “simple” representation theorem.

Theorem 8. Assume that $X \equiv \mathbb{R}_+^L$ and $\succsim$ is a monotone and continuous preference relation on X . Then, $\succsim$ admits a continuous utility representation.

This proof requires the following mathematical fact: $\mathbb{R}_+$ is connected, in particular, if A and B are non-empty closed subsets of $\mathbb{R}_+$ such that $A \cup B = \mathbb{R}$ then $A \cap B \neq \emptyset$.

The strategy of the proof is to show that for each $x \in \mathbb{R}_+^L$ there is a unique $a \in \mathbb{R}_+$ such that $(a, \dots, a) \sim x$. This number is a welfare index for x .

Proof. Let $x \in \mathbb{R}_+^L$. We prove that there is $a \in \mathbb{R}_+$ such that $(a, \dots, a) \sim x$. Let $A \equiv \{a \in \mathbb{R}_+ : (a, \dots, a) \succsim x\}$ and $B \equiv \{b \in \mathbb{R}_+ : x \succsim (b, \dots, b)\}$.

- $A \neq \emptyset$. Let $a \equiv \max_{l=1, \dots, L} x_l$. Since $\succsim$ is monotone, then $(a+1, \dots, a+1) \gg x$ implies that $(a+1, \dots, a+1) \succsim x$. Thus, $a+1 \in A$.

- $B \neq \emptyset$. We prove that $0 \in B$. For each $n \in \mathbb{N}$, let $x_n = (x_1 + \frac{1}{n}, \dots, x_L + \frac{1}{n})$. As $n \rightarrow \infty$, $x_n \rightarrow x$. Let $n \in \mathbb{N}$. Since $x_n \gg 0$, then by monotonicity of $\succsim$, $x_n \succ 0$. By continuity of $\succsim$, $\lim_{n \rightarrow \infty} x_n = x \succsim 0$.

- $A \cup B = \mathbb{R}_+$. Clearly $A \cup B \subseteq \mathbb{R}_+$. Let $a \in \mathbb{R}_+$. Since $\succsim$ is complete, then either $x \succsim (a, \dots, a)$ or $(a, \dots, a) \succ x$. Thus, either $a \in A$ or $a \in B$.

- A and B are closed. We prove that A is closed. Let $\{a_n\}_{n \in \mathbb{N}}$ be a sequence in A such that as $n \rightarrow \infty$, $a_n \rightarrow a \in \mathbb{R}_+$. Let $n \in \mathbb{N}$. Since $a_n \in A$,

then $(a_n, \dots, a_n) \succsim x$. Thus, by continuity of $\succsim$, $\lim_{n \rightarrow \infty} a_n = a \succsim x$. Thus, $a \in A$.

Since $A \cup B = \mathbb{R}_+$, A and B are non-empty closed sets, then $A \cap B \neq \emptyset$. Let $a \in A \cap B$. Thus, $(a, \dots, a) \succsim x \succsim (a, \dots, a)$. Thus, $(a, \dots, a) \sim x$.

We claim that for each $x \in \mathbb{R}_+^L$, there is a unique $a \in \mathbb{R}_+$ such that $(a, \dots, a) \sim x$. Let $\{a, a'\} \subseteq \mathbb{R}_+$ be such that $(a, \dots, a) \sim x$ and $(a', \dots, a') \sim x$. Suppose by means of contradiction that $a > a'$. By monotonicity of $\succsim$, $(a, \dots, a) \succ (a', \dots, a')$. By transitivity of $\succsim$, $x \succ x$. This contradicts completeness of $\succsim$. Let u be the function defined by

$$\begin{aligned} u : \mathbb{R}_+^L &\rightarrow \mathbb{R} \\ x &\mapsto u(x) \text{ such that } (u(x), \dots, u(x)) \sim x. \end{aligned}$$

One can prove that u represents $\succsim$ (see Problem 9, Section 3.3.3).

It can be proved that u is a continuous function... $\square$

The following theorem states more general conditions for the existence of a continuous representation of a preference. Indeed, it is enough to assume separability of the alternatives space X . Since Euclidean spaces are separable, then a continuous preference in a consumption space is always representable by a continuous function.

Theorem 9 (Debreu, 1963). Let $\succsim$ be a preference on $X \subseteq \mathbb{R}_+^L$. If $\succsim$ is continuous, then there is a continuous function $u : X \rightarrow \mathbb{R}$ representing $\succsim$.

3.3.2 Further requirements on preferences

Convex preferences

Definition 56. A function $u : \mathbb{R}^L \rightarrow \mathbb{R}$ is **quasi-concave** if for each $\{x, y\} \subseteq \mathbb{R}^L$ and each $\alpha \in [0, 1]$, $u(\alpha x + (1 - \alpha)y) \geq \min\{u(x), u(y)\}$.

Definition 57. A function $u : \mathbb{R}^L \rightarrow \mathbb{R}$ is **strictly quasi-concave** if for each $\{x, y\} \subseteq \mathbb{R}^L$ such that $x \neq y$ and each $\alpha \in (0, 1)$, $u(\alpha x + (1 - \alpha)y) > \min\{u(x), u(y)\}$.

Proposition 8. Let $\succsim$ be a preference on $\mathbb{R}_+^L$ that is represented by a utility function u . Then, $\succsim$ is convex if and only if u is quasi-concave. Moreover, $\succsim$ is strictly convex if and only if u is strictly quasi-concave.

Proof. See Problem 7, Section 3.3.3. $\square$

Homothetic preferences

The following theorem states that continuous homothetic preferences are represented by functions that are homogeneous of degree one. Observe that no monotonicity condition is required for the existence of such a representation.

Theorem 10. Assume $X \equiv \mathbb{R}_+^L$. A continuous preference $\succsim$ is homothetic if and only if it admits a continuous representation $u : X \rightarrow \mathbb{R}$ that is homogeneous of degree one, i.e., such that for each $\alpha \in \mathbb{R}_+$, $u(\alpha x) = \alpha u(x)$.

Proof. See Problems 2 and 3, Section 3.3.3. $\square$

Quasilinear preferences

Proposition 9. A continuous preference relation $\succsim$ on $X \equiv \mathbb{R} \times \mathbb{R}_+^{L-1}$ is quasi-linear with respect to the first commodity if and only if it admits a continuous utility representation u of the form: for each $x \in X$, $u(x) \equiv x_1 + \varphi(x_2, \dots, x_L)$.

Proof. See Problem 8, Section 3.3.3. $\square$

3.3.3 Problems

1. Prove that $\succsim_{lex}$ is not continuous.
2. Assume $X \equiv \mathbb{R}_+^L$. Prove that a monotone and continuous preference $\succsim$ is homothetic if and only if it admits a representation $u : X \rightarrow \mathbb{R}$ that is homogeneous of degree one, i.e., such that for each $\alpha \in \mathbb{R}_+$, $u(\alpha x) = \alpha u(x)$. Hint: ($\Rightarrow$) For each $x \in X$ let $u(x) \in \mathbb{R}_+$ be such that $x \sim (u(x), \dots, u(x))$. Show that u is a well defined function, i.e., for each x there is always such a real number and it is unique. Prove that this function is homogeneous of degree one.
3. (Additional) Prove Theorem 10 (the difference with Problem 2 is that no monotonicity condition is assumed here).
4. Construct a continuous and homothetic preference on $\mathbb{R}_+^2$ that is not monotone.
5. Assume for simplicity that $X \equiv \mathbb{R}_+^2$. Prove that for each homothetic preference $\succsim$, the slope of the indifference sets is unchanged along rays through the origin (assume differentiability). Now, assume that the

preference is also convex. Generalize and prove the slope statement to a statement concerning constant supporting planes.

6. Prove that Cobb-Douglas preferences on $\mathbb{R}_+^2$ are homothetic and monotone, but not strictly monotone.
7. Prove Proposition 8.
8. Prove Proposition 9.
9. Complete the proof of Theorem 8. Prove that the function u represents $\succsim$ (you don't have to prove that u is continuous).
10. Consider an L -commodity space $\mathbb{R}_+^L$ and a preference $\succsim$ on $\mathbb{R}_+^L$.
 - (a) State the definition of homothetic preferences.
 - (b) State the definition of continuous preferences.
 - (c) Prove that if preferences are homothetic and continuous, then for each $\{x, y\} \subseteq \mathbb{R}_+^L$ such that $x \succsim y$, and each $\alpha > 0$, $\alpha x \succsim \alpha y$.
11. Let $\succsim$ be a monotone and continuous preference on $\mathbb{R}_+^L$. Let $p \in \mathbb{R}_{++}^L$. Prove that $\succsim$ is represented by the function m_p that associates with each $x \in \mathbb{R}_+^L$ the value

$$m_p(x) \equiv \min_{y \in U(\succsim, x)} p \cdot y.$$

Do it in three steps:

- (a) Write the statement “ m_p represents $\succsim$ ” as an if and only if statement.
- (b) Provide an explicit proof of $\Rightarrow$.
- (c) Provide an explicit proof of $\Leftarrow$.

3.4 Consumer theory III: Choice

In this section we investigate agents' choice in a competitive environment.

3.4.1 Competitive environment

- The set of alternatives X has the same structure as in the consumer's preference model, i.e., $X \subseteq \mathbb{R}_+^L$. We will assume for simplicity that $X \equiv \mathbb{R}_+^L$.
- The competitive (Walrasian) budget set at prices $p \in \mathbb{R}_+^L$ and wealth $w \in \mathbb{R}_+$ is the set $B(p, w) \equiv \{x \in X : p \cdot x \leq w\}$.
- We consider agents' choices from the family of competitive (Walrasian) budget sets $\mathcal{B}^W \equiv \{B(p, w) : p \in \mathbb{R}_+^L, w \in \mathbb{R}_+\}$.
- An agent in a competitive environment is represented by a Walrasian choice structure $(\mathcal{B}^W, C)$. Choice in this environment is interpreted as demand. Thus, we refer to C as the agent's demand correspondence. In order to simplify notation we write $C(p, w)$ instead of $C(B(p, w))$.
- Choice structure $(\mathcal{B}, C)$ is single-valued if for each $B \in \mathcal{B}$, $C(B)$ is a singleton. Each single-valued choice structure induces a demand function. The demand function associated with a single-valued choice structure $(\mathcal{B}^W, C)$, denoted x , is given by: for each $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$, $x(p, w)$ is the unique element in $C(p, w)$.

In what follows, whenever we refer to a choice structure, we implicitly assume that it is a competitive choice structure, i.e., $(\mathcal{B}^W, C)$. Whenever we refer to a demand function, we implicitly assume that there is a single-valued choice structure that generates it.

Definition 58. A function $f : \mathbb{R}_+^M \rightarrow \mathbb{R}$ is **homogeneous of degree** $r \in \mathbb{N}$, if for each $\alpha \in \mathbb{R}_+$ and each $z \in \mathbb{R}_+^M$, $f(\alpha z) = \alpha^r f(z)$.

Remark 6. The demand function associated with a single-valued Walrasian choice structure is homogeneous of degree zero.

Are there sensible restrictions on choice if we interpret commodities as “goods”?

Definition 59. A choice structure $(\mathcal{B}^W, C)$ satisfies Walras' law if for each $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$ and each $x \in C(p, w)$, $p \cdot x = w$. A demand function x satisfies Walras' law if for each $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$, $p \cdot x(p, w) = w$.

3.4.2 Comparative statics

What are the changes in consumption induced by a change in wealth and prices?

Wealth effects

Definition 60. Let x be a demand function and $p \in \mathbb{R}_+^L$. The **wealth expansion path**² (or **Engle function**) for x at p is the function: $w \in \mathbb{R}_+ \mapsto x(p, w)$. We usually denote this function $x(p, \cdot)$.

Definition 61. Demand x is **normal** if for each $p \in \mathbb{R}_+^L$ the wealth expansion path for x at p is non-decreasing.

Definition 62. Let x be a differentiable demand function. Demand for commodity l is **normal at** $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$, if $\frac{\partial x_l}{\partial w}(p, w) \geq 0$.

Definition 63. Let x be a differentiable demand function. The **matrix of wealth effects at** $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$ is the partial derivative of x with respect to wealth, ie.,

$$D_w x(p, w) \equiv \begin{bmatrix} \frac{\partial x_1}{\partial w}(p, w) \\ \vdots \\ \frac{\partial x_L}{\partial w}(p, w) \end{bmatrix}$$

Price effects

Definition 64. Let x be a differentiable demand function. Commodity l is a **“Giffen good” at** $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$, if $\frac{\partial x_l}{\partial p_l}(p, w) > 0$.

Definition 65. Let x be a differentiable demand function. The **matrix of price effects at** $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$ is the matrix of partial derivatives of x with respect to prices, ie.,

$$D_p x(p, w) \equiv \begin{bmatrix} \frac{\partial x_1}{\partial p_1}(p, w) & \dots & \frac{\partial x_1}{\partial p_L}(p, w) \\ \vdots & & \vdots \\ \frac{\partial x_L}{\partial p_1}(p, w) & \dots & \frac{\partial x_L}{\partial p_L}(p, w) \end{bmatrix}$$

²Mas-Colell et al. defines wealth expansion path as the graph of the Engle function, i.e., $\{x(p, w) : w > 0\}$.

Elasticities

Definition 66. Let x be a differentiable demand function. The **elasticity of demand for commodity l with respect to wealth at** $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$, is

$$\varepsilon_{lw}(p, w) \equiv \frac{\partial x_l}{\partial w}(p, w) \frac{w}{x_l(p, w)}.$$

Definition 67. Let x be a differentiable demand function. Commodity l is a **luxury for $\succsim$ at** $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$, if the elasticity of demand for commodity l with respect to wealth at (p, w) is greater than or equal to 1.

Definition 68. Let x be a differentiable demand function. The **elasticity of demand for commodity l with respect to price of commodity k at** $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$, is

$$\varepsilon_{lk}(p, w) \equiv \frac{\partial x_l}{\partial p_k}(p, w) \frac{p_k}{x_l(p, w)}.$$

3.4.3 WARP

The following proposition states that a single valued choice structure satisfies WARP if and only if for each two different budget sets in which the choices are different, the choices cannot be both in the intersection of the sets.

Proposition 10. A single-valued Walrasian choice structure $(\mathcal{B}^W, C)$, satisfies WARP if and only if its associated demand function x satisfies the following property: for any two price-wealth combinations, (p, w) and (p', w') , if $[p \cdot x(p', w') \leq w$ and $x(p', w') \neq x(p, w)]$, then $[p' \cdot x(p, w) > w']$.

Proof. ($\Rightarrow$) Suppose that the choice structure is single-valued with demand function x and satisfies WARP. We want to prove that for each pair of price-wealth configurations (p, w) and (p', w') , if $[p \cdot x(p', w') \leq w$ and $x(p', w') \neq x(p, w)]$, then $[p' \cdot x(p, w) > w']$. So suppose that $p \cdot x(p', w') \leq w$ and $x(p', w') \neq x(p, w)$. We need to prove that $p' \cdot x(p, w) > w'$. Suppose by contradiction that $p' \cdot x(p, w) \leq w'$. Let $A \equiv B(p, w)$ and $B \equiv B(p', w')$. Then, $\{A, B\} \subseteq \mathcal{B}^W$, $x(p, w) \in C(A)$, $x(p', w') \in C(B)$. Since $p \cdot x(p', w') \leq w$, then $x(p', w') \in A$. Since $p' \cdot x(p, w) \leq w'$, then $x(p, w) \in B$. By definition $x(p, w) \in A$ and $x(p', w') \in B$. Since $(\mathcal{B}^W, C)$ satisfies WARP, then $x(p', w') \in C(A)$. Since $x(p', w') \neq x(p, w)$, then $C(A)$ is not single valued. This is a contradiction.

($\Leftarrow$) Suppose that the choice structure is single-valued with demand function x , and that for each pair of price-wealth configurations (p, w) and

(p', w') , if $[p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w)]$, then $[p' \cdot x(p, w) > w']$. We prove that $(\mathcal{B}^W, C)$ satisfies WARP. Suppose by contradiction that there are $\{A, B\} \subseteq \mathcal{B}^W$, $\{x, y\} \in A \cap B$ such that $x \in C(A)$, $y \in C(B)$, and $x \notin C(B)$. Since $x \notin C(B)$, then $x \neq y$. Since $\{A, B\} \subseteq \mathcal{B}^W$, then there are (p, w) and (p', w') such that $A = B(p, w)$ and $B = B(p', w')$. Since $x \in C(A)$, then $x(p, w) = x$. Since $y \in C(B)$, then $x(p', w') = y$. Since $y \in A$, then $p \cdot x(p', w') \leq w$. Since $p \cdot x(p', w') \leq w$ and $x(p, w) \neq x(p', w')$, then $p' \cdot x(p, w) > w'$. Thus, $x = x(p, w) \notin B$. This is a contradiction. $\square$

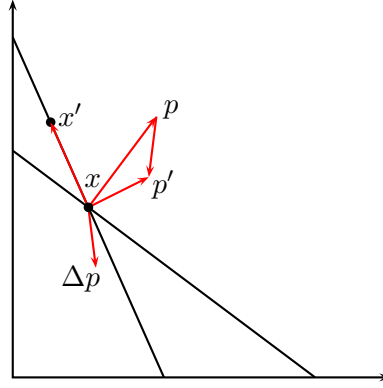


Figure 3.6: Compensated law of demand.

Proposition 11 (Compensated law of demand). A single-valued Walrasian choice structure that satisfies Walras' law satisfies WARP if and only if its associated demand function x satisfies the following property: for each $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$ and each $p' \in \mathbb{R}_+^L$,

$$[p' - p]^T [x(p', w') - x(p, w)] \leq 0,$$

where $w' \equiv p' \cdot x(p, w)$, and this inequality is strict whenever $x(p', w') \neq x(p, w)$.

Proof. Suppose that $(\mathcal{B}^W, x)$ satisfies WARP. Suppose that $x(p, w) \neq x(p', w')$. Since $w' = p' \cdot x(p, w)$, then

$$[p' - p]^T [x(p', w') - x(p, w)] = p^T x(p, w) - p^T x(p', w').$$

Thus, it is enough to prove that $p^T x(p, w) < p^T x(p', w')$. Suppose by means of contradiction that $p^T x(p, w) \geq p^T x(p', w')$. By Proposition 10, $p' \cdot x(p, w) > w'$. Thus, $w' > w'$. This is a contradiction. $\square$

3.4.4 Problems

1. Prove Propositions 2.E.1, 2.E.2, 2.E.3, and Exercise 2.E.2 in [Mas-Colell et al. \(1995\)](#).
2. Consider a Cobb-Douglass preference represented by the utility function

$$U(x_1, x_2) \equiv x_1^\alpha x_2^{1-\alpha}.$$

- (a) Show that the preference is convex.
- (b) Show that for each $p \in \mathbb{R}_{++}^2$ and each $w > 0$, there is a unique maximizer for U in the budget set $B(p, w)$.
- (c) Define the demand function associated to U as the function $(p, w) \in \mathbb{R}_{++}^2 \times \mathbb{R}_+ \mapsto x(p, w) \in \mathbb{R}_+^2$ where $x(p, w)$ is the unique maximizer of U in $B(p, w)$. Find an explicit expression for $x(p, w)$.
- (d) State the compensated law of demand for a demand function. Show that the demand function associated with U satisfies the Compensated Law of demand.

The demand function associated with this preference is the function given by:

3. Consider the preference on $\mathbb{R}_+^2$ represented by

$$U(x, y) = (x + a)^\alpha (y + b)^{1-\alpha},$$

where $a > 0$ and $b > 0$. The demand function associated with this preference is given by:

- (a) Illustrate income expansion paths for the demand function associated with this preference.
- (b) Provide an expression for the demand function associated with this preference when consumption is interior.
- (c) A good is said to be a luxury good at (p, w) if its income elasticity is greater than or equal to 1. Provide a characterization in terms of $\frac{\partial U}{\partial x}(0, 0)$ and $\frac{\partial U}{\partial y}(0, 0)$ of the combinations (p, w) for which a good, say x , is a luxury good when its consumption is interior.

3.4.5 Differential characterization of WARP

Definition 69. Let $(\mathcal{B}^W, C)$ be a single-valued Walrasian choice structure and x its associated demand function. The **Slutsky compensated demand of x at $(\bar{p}, \bar{w}) \in \mathbb{R}_+^L \times \mathbb{R}_+$** is the function

$$x^{(\bar{p}, \bar{w})} : \mathbb{R}_+^L \rightarrow \mathbb{R}_+^L$$

$$p \mapsto x^{(\bar{p}, \bar{w})}(p) \equiv x(p, p \cdot x(\bar{p}, \bar{w})) \equiv \begin{bmatrix} x_1(p, p \cdot x(\bar{p}, \bar{w})) \\ \vdots \\ x_L(p, p \cdot x(\bar{p}, \bar{w})) \end{bmatrix}$$

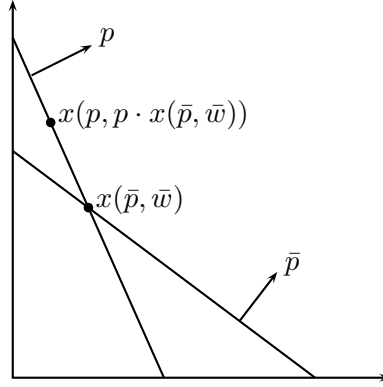


Figure 3.7: Slutsky compensated law of demand.

Proposition 12. Let $(\mathcal{B}^W, C)$ be a single-valued Walrasian choice structure and x its associated demand function. If x is differentiable, then for each $(\bar{p}, \bar{w}) \in \mathbb{R}_+^L \times \mathbb{R}_+$, $x^{(\bar{p}, \bar{w})}$ is differentiable and for each $p \in \mathbb{R}_+^L$ and each pair $\{l, k\} \subseteq \{1, \dots, L\}$,

$$\frac{\partial x_l^{(\bar{p}, \bar{w})}}{\partial p_k}(p, p \cdot x(\bar{p}, \bar{w})) = \frac{\partial x_l}{\partial p_k}(p, p \cdot x(\bar{p}, \bar{w})) + \frac{\partial x_l}{\partial w}(p, p \cdot x(\bar{p}, \bar{w}))x_k(p, p \cdot x(\bar{p}, \bar{w}))$$

or in matrix notation,

$$D_p x^{(\bar{p}, \bar{w})}(p) = D_p x(p, p \cdot x(\bar{p}, \bar{w})) + D_w x(p, p \cdot x(\bar{p}, \bar{w}))x(\bar{p}, \bar{w})^T.$$

Proof. Straightforward application of the chain rule given that

$$x_l^{(\bar{p}, \bar{w})}(p) = x_l(p_1, \dots, p_L, p_1 x_1(\bar{p}, \bar{w}) + \dots + p_L x_L(\bar{p}, \bar{w})).$$

□

Definition 70. Let $(\mathcal{B}^W, C)$ be a single-valued Walrasian choice structure that satisfies Walras' law and such that its associated demand function x is differentiable. The derivative of the Slutsky compensated demand of x at $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$, evaluated at p , denoted $S(p, w)$, is called the **substitution**, or **Slutsky**, matrix of $(\mathcal{B}^W, C)$ at (p, w) . That is, for each $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$ and each pair $\{l, k\} \subseteq \{1, \dots, L\}$,

$$S_{lk}(p, w) = \frac{\partial x_l}{\partial p_k}(p, w) + \frac{\partial x_l}{\partial w}(p, w)x_k(p, w),$$

or in matrix notation,

$$S(p, w) = D_p x(p, w) + D_w x(p, w)x(p, w)^T.$$

Proposition 13. Let $(\mathcal{B}^W, C)$ be a single-valued Walrasian choice structure that satisfies WARP, Walras' law, and its associated demand function x is twice differentiable. Then for each $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$ and each $v \in \mathbb{R}^L$,

$$v^T S(p, w) v \leq 0.$$

Proof. Let $\varepsilon > 0$ and $p^\varepsilon \equiv p + \varepsilon v$. Since x is C^1 , then by Taylor's formula

$$x^{(p, w)}(p^\varepsilon) - x^{(p, w)}(p) = D_p x^{(p, w)}(p)(p^\varepsilon - p) + R(p, p^\varepsilon) = S(p, w)(p^\varepsilon - p) + R(p, p^\varepsilon),$$

where $\lim_{p^\varepsilon \rightarrow p} \frac{R(p, p^\varepsilon)}{\|p^\varepsilon - p\|} = 0$.

Since $(\mathcal{B}^W, C^{\succsim})$ satisfies Walras' law, then $x^{(p, w)}(p) = x(p, w)$, so

$$x^{(p, w)}(p^\varepsilon) - x^{(p, w)}(p) = x(p^\varepsilon, p^\varepsilon \cdot x(p, w)) - x(p, w).$$

Thus,

$$x(p^\varepsilon, p^\varepsilon \cdot x(p, w)) - x(p, w) = S(p, w)(p^\varepsilon - p) + R(p, p^\varepsilon),$$

and

$$[p^\varepsilon - p]^T [x(p^\varepsilon, p^\varepsilon \cdot x(p, w)) - x(p, w)] = [p^\varepsilon - p]^T [S(p, w)(p^\varepsilon - p) + R(p, p^\varepsilon)].$$

By the compensated law of demand,

$$[p^\varepsilon - p]^T [S(p, w)[p^\varepsilon - p] + R(p, p^\varepsilon)] \leq 0,$$

Recall that $p^\varepsilon - p = \varepsilon v$. Thus,

$$\varepsilon v^T S(p, w) [\varepsilon v] + \varepsilon v^T R(p, p^\varepsilon) \leq 0.$$

Dividing by ε^2 and rearranging terms,

$$v^T S(p, w) v \leq -\frac{v^T R(p, p^\varepsilon)}{\varepsilon}.$$

Since the expression above holds for each $\varepsilon > 0$ then (whenever the limit of the expression on the right exists)

$$v^T S(p, w) v \leq \lim_{\varepsilon \rightarrow 0} -\frac{v^T R(p, p^\varepsilon)}{\varepsilon}.$$

Finally, since $\|p^\varepsilon - p\| = \varepsilon\|v\|$, then

$$\lim_{\varepsilon \rightarrow 0} -\frac{v^T R(p, p^\varepsilon)}{\varepsilon} = -\|v\| v^T \lim_{\varepsilon \rightarrow 0} \frac{R(p, p^\varepsilon)}{\|p^\varepsilon - p\|} = 0.$$

Thus,

$$v^T S(p, w) v \leq 0.$$

□

Corollary 2. If a single-valued choice structure satisfies Walras' law and WARP and its associated demand function is twice differentiable, then its Slutsky matrix is negative semi-definite.

Proposition 14. Let $(\mathcal{B}^W, C)$ be a single-valued choice structure that satisfies Walras' law and its associated demand function x is differentiable. Then for each $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$, $p^T S(p, w) = 0$ and $S(p, w)p = 0$.

Proof. Problem 1, Section 3.4.8.

□

Corollary 3. Let $(\mathcal{B}^W, C)$ be a single-valued choice structure that satisfies Walras' law and its associated demand function x is twice differentiable. Let S_l be the matrix obtained from $S(p, w)$ by deleting the l -th row and l -th column for some $l \in \{1, \dots, L\}$. If $(\mathcal{B}^W, C)$ satisfies WARP, then the symmetric matrix $S_l + S_l^T$ is negative semi-definite.

Proposition 15. Let $(\mathcal{B}^W, C)$ be a single-valued choice structure that satisfies Walras' law and its associated demand function x is twice differentiable. Let S_l be the matrix obtained from $S(p, w)$ by deleting the l -th row and l -th column for some $l \in \{1, \dots, L\}$. If the symmetric matrix $S_l + S_l^T$ is negative definite, then $(\mathcal{B}^W, C)$ satisfies WARP.

Proof. Problem 2, Section 3.4.8.

□

3.4.6 Strong Axiom of Revealed Preference: SARP

Let (B^W, C) be a single-valued choice structure that satisfies Walras' law and WARP. Are there preferences on X that rationalize such a choice structure? Not necessarily (this is possible because B^W does not satisfy the pair-and-triple condition). This will be clear after characterizing the choice structures induced by maximization of preferences.

The following stronger consistency condition guarantees that a single-valued choice structure that satisfies Walras' law can be rationalized by a preference relation.

Definition 71. *The demand function x satisfies the **Strong Axiom of Revealed Preference (SARP)** if for each list $(p^1, w^1), \dots, (p^M, w^M)$, such that for each $m \in \{1, \dots, M-1\}$, $p^m \cdot x(p^{m+1}, w^{m+1}) \leq w^m$ and $x(p^{m+1}, w^{m+1}) \neq x(p^m, w^m)$, we have that $p^M \cdot x(p^1, w^1) > w^M$.*

Proposition 16. WARP implies SARP when there are exactly two commodities (in our notation $L = 2$).

Proof. Problem 8, Section 3.4.8. □

3.4.7 Matrix Definiteness

In this section I collect some definitions and characterizations of “matrix definiteness.” (Some other test that characterize definiteness are available in the literature; of special importance are those that relate the eigenvalues of a matrix with its definiteness properties).

Definition 72. *An $n \times n$ matrix A is said to be*

1. **positive definite** if for each $x \in \mathbb{R}^n$, $x^T A x > 0$.
2. **positive semi-definite** if for each $x \in \mathbb{R}^n$, $x^T A x \geq 0$.
3. **negative definite** if for each $x \in \mathbb{R}^n$, $x^T A x < 0$.
4. **negative semi-definite** if for each $x \in \mathbb{R}^n$, $x^T A x \leq 0$.

Lemma 2. An $n \times n$ matrix A is

1. positive definite if and only if the symmetric matrix $A + A^T$ is positive definite.
2. positive semi-definite if and only if the symmetric matrix $A + A^T$ is positive semi-definite.

3. negative definite if and only if the symmetric matrix $A + A^T$ is negative definite.
4. negative semi-definite if and only if the symmetric matrix $A + A^T$ is negative semi-definite.

Proof. For each $x \in \mathbb{R}^n$, $x^T(A + A^T)x = 2x^T Ax$. □

Identifying Definiteness

Definition 73. Let A_k be a submatrix of an $n \times n$ matrix A obtained when only the first k rows and columns are retained:

$$A_k = \begin{bmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}.$$

Then, A_k is called the k -th naturally ordered principal minor of A .

Theorem 11 (Sylvester's criterion). Let A be a symmetric $n \times n$ matrix. Then,

1. A is negative definite $\Leftrightarrow$

$$\text{for each } k \in \{1, \dots, n\}, (-1)^k |A_k| > 0,$$

where $|A_k|$ denotes the determinant of square matrix A_k .

2. A is positive definite $\Leftrightarrow$

$$\text{for each } k \in \{1, \dots, n\}, |A_k| > 0,$$

where $|A_k|$ denotes the determinant of square matrix A_k .

Definition 74. π is a permutation if it is a one-to-one function from $\{1, \dots, n\}$ onto $\{1, \dots, n\}$.

- Given an $n \times n$ matrix and a permutation $\pi = (\pi_1, \dots, \pi_n)$, define

$$A^\pi = \begin{bmatrix} a_{\pi_1 \pi_1} & \cdots & a_{\pi_1 \pi_n} \\ \vdots & a_{\pi_i \pi_j} & \vdots \\ a_{\pi_n \pi_1} & \cdots & a_{\pi_n \pi_n} \end{bmatrix}.$$

A^π is another $n \times n$ matrix obtained by applying the permutation π to both the rows and the columns of A .

- Example: Let $\pi = (\pi_1 \ \pi_2 \ \pi_3) = (2 \ 3 \ 1)$ and

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix},$$

then

$$A^\pi = \begin{bmatrix} a_{\pi_1\pi_2} & a_{\pi_1\pi_3} & a_{\pi_1\pi_3} \\ a_{\pi_2\pi_2} & a_{\pi_2\pi_3} & a_{\pi_2\pi_3} \\ a_{\pi_3\pi_2} & a_{\pi_3\pi_3} & a_{\pi_3\pi_3} \end{bmatrix} = \begin{bmatrix} a_{22} & a_{23} & a_{21} \\ a_{32} & a_{33} & a_{31} \\ a_{12} & a_{13} & a_{11} \end{bmatrix} = \begin{bmatrix} 5 & 6 & 4 \\ 8 & 9 & 7 \\ 2 & 3 & 1 \end{bmatrix}.$$

Theorem 12 (Sylvester's criterion). Let A be a symmetric $n \times n$ matrix. Then,

1. A is positive semidefinite $\Leftrightarrow$

for each $k \in \{1, \dots, n\}$, and for each possible permutation π , $|A_k^\pi| \geq 0$,

where $|A_k|$ denotes the determinant of square matrix A_k .

2. A is negative semidefinite $\Leftrightarrow$

for each $k \in \{1, \dots, n\}$, and for each possible permutation π , $(-1)^k |A_k^\pi| \geq 0$,

where $|A_k|$ denotes the determinant of square matrix A_k .

3.4.8 Problems

1. Prove Propositions 2.F.3 in [Mas-Colell et al. \(1995\)](#).
2. (Additional) Prove Proposition [15](#).
3. Let $X \equiv \mathbb{R}^3$. Suppose that a Walrasian demand function x satisfies Walras' law and is such that for each pair $(p, w) \in \mathbb{R}_{++}^3 \times \mathbb{R}_+$:

$$x_1(p, w) \equiv a + b \frac{p_1}{p_3} + c \frac{p_2}{p_3},$$

$$x_2(p, w) \equiv d + e \frac{p_1}{p_3} + f \frac{p_2}{p_3}.$$

- (a) Calculate x_3 .
- (b) Is x homogeneous of degree zero?

- (c) Provide restrictions on the demand parameters that guarantee the associated choice structure satisfies WARP. (Disregard the fact that our results were developed under the assumption $X \equiv \mathbb{R}_+^L$)
4. Let $X \equiv \mathbb{R}^3$ be the set of alternatives. Suppose that a demand function x satisfies Walras' law and is such that for each pair $(p, w) \in \mathbb{R}_{++}^3 \times \mathbb{R}_+$:

$$x_1(p, w) \equiv 100 + \frac{1}{10} \frac{p_1}{p_3} + 20 \frac{p_2}{p_3},$$

$$x_2(p, w) \equiv 450 + 20 \frac{p_1}{p_3} - \frac{5}{3} \frac{p_2}{p_3}.$$

- (a) Calculate $x_3(p, w)$.
- (b) Is x homogeneous of degree zero?
- (c) Write the lk component of the Slutsky matrix associated with x , $S_{lk}(p, w)$, in terms of the partial derivatives of x . Calculate $S_{11}(p, w)$, $S_{12}(p, w)$, $S_{21}(p, w)$, and $S_{22}(p, w)$.
- (d) Is x the result of the maximization of preferences?
5. (Additional) Let $(\mathcal{B}^W, C)$ be a single-valued choice structure that satisfies Walras' law and its associated demand function x is C^1 . Let S_l be the matrix obtained from $S(p, w)$ by deleting the l -th row and l -th column for some $l \in \{1, \dots, L\}$. Prove that $S(p, w)$ is negative semi-definite if and only if the symmetric matrix $S_l + S_l^T$ is negative semi-definite?
6. Consider a two-commodity consumption space $\mathbb{R}_+^2$. The generic demand function (or equivalently, a single-valued correspondence) is denoted x and associates with each combination of prices and wealth a consumption bundle, i.e.,

$$(p, w) \in \mathbb{R}_{++}^2 \times \mathbb{R}_+ \mapsto x(p, w) \equiv (x_1(p, w), x_2(p, w)).$$

- (a) Prove that WARP is equivalent to the following property for demand functions in a competitive environment:
For each pair $\{p, p'\} \subseteq \mathbb{R}_{++}^2$ and each $\{w, w'\} \subseteq \mathbb{R}_+$, if $p' \cdot x(p, w) \leq w'$ and $x(p, w) \neq x(p', w')$, then $p \cdot x(p', w') > w$.
- (b) Prove that if x is a demand function that is homogeneous of degree zero and satisfies Walras' Law, then WARP is equivalent to the following property (Hint: use (a)).
For each pair $\{p, p'\} \subseteq \mathbb{R}_{++}^2$, if $p' \cdot x(p, 1) < 1$, then $p \cdot x(p', 1) > 1$.

7. Consider a two-commodity consumption space $\mathbb{R}_+^2$ in which the generic demand function is denoted x . We say that demand function x satisfies the **weak gross substitutes property (WGSP)** if the following two conditions are satisfied:

- For each $p_1 \in \mathbb{R}_{++}$ and each $w \in \mathbb{R}_+$, the function $p_2 \mapsto x_1(p_1, p_2, w)$ is non-decreasing;
- For each $p_2 \in \mathbb{R}_{++}$ and each $w \in \mathbb{R}_+$, the function $p_1 \mapsto x_2(p_1, p_2, w)$ is non-decreasing.

Assume that x is homogeneous of degree zero, satisfies Walras' law and WGSP.

- (a) Let $\{p, p'\} \subseteq \mathbb{R}_{++}^2$ be such that $p'_1 \geq p_1$ and $p'_2 \leq p_2$. Prove that $x_1(p', 1) \leq x_1(p, 1)$ and $x_2(p', 1) \geq x_2(p, 1)$. (Hint: First argue that $x_1(p'_1, p'_2, 1) \leq x_1(p'_1, p_2, 1)$. Then argue that $x_1(p'_1, p_2, 1) \leq x_1(p_1, p_2, 1)$.)
 - (b) Prove that if x satisfies WGSP then it satisfies WARP. (Hint: use the statement proved in Problem 6 (b), Section 3.4.8; suppose that $p' \cdot x(p, 1) < 1$, then prove that $p \cdot x(p', 1) > 1$; analyze three cases: (i) $p' \geq p$, (ii) $p'_2 < p_2$ and $p'_1 < p_1$, and (iii) $p'_2 < p_2$ and $p'_1 \geq p_1$ -here it is useful to observe that $\Delta p \cdot \Delta x = \Delta p_1 \cdot \Delta x_1 + \Delta p_2 \cdot \Delta x_2$ and also $\Delta p \cdot \Delta x = p' \cdot x' - p' \cdot x - p \cdot x' + p \cdot x$).
 - (c) Does WARP imply WGSP?
8. (Additional) Prove Proposition 16.

3.5 Consumer Theory IV: Revealed Preference

What are the testable implications of consumers' preference maximization? For simplicity, we assume throughout this section that prices are strictly positive. We denote the set of Walrasian budget sets with strictly positive prices $\mathcal{B}_{++}^W$. We assume throughout that $X \equiv \mathbb{R}_+^L$.

3.5.1 Preference maximization

Let $\succsim$ be a preference relation on X . Recall from our abstract study of choice that the choices of the agent represented by $\succsim$ in a budget set $B(p, w) \in \mathcal{B}^W$ are given by:

$$C^{\succsim}(p, w) \equiv \{x \in B(p, w) : \text{for each } y \in B(p, w), x \succsim y\}.$$

3.5.2 Utility maximization

Suppose that preference $\succsim$ is represented by utility function u . Then $x^* \in C^{\succsim}(p, w)$ if and only if for each $y \in B(p, w)$, $u(x) \geq u(y)$. So x^* is chosen by agent $\succsim$ if and only if it solves the **Utility Maximization Problem (UMP)**:

$$\begin{aligned} & \max u(x) \\ & \text{subject to } x \in \mathbb{R}_+^L \text{ and } p \cdot x \leq w \end{aligned}$$

Definition 75. Let A be a set and $f : A \rightarrow \mathbb{R}$ a real valued function defined on A . An element $x \in A$ is a **maximizer of f on A** if for each $y \in A$, $f(x) \geq f(y)$. The set of maximizers of f on A is denoted **arg max** $_{x \in A} f(x)$ or alternatively **arg max** $\{f(x) : x \in A\}$.

Using the notation above, for each $(p, w) \in \mathbb{R}_+^L \times \mathbb{R}_+$,

$$C^{\succsim}(p, w) = \arg \max \{u(x) : x \in B(p, w)\}.$$

3.5.3 Characterization of demand correspondence

Proposition 17. Let $\succsim$ be a continuous preference on X . Then, for each $p \gg 0$ and each $w \in \mathbb{R}_+$, $C^{\succsim}(p, w) \neq \emptyset$.

Proof. Continuous preferences on $X \subseteq \mathbb{R}_+^L$ admit a continuous representation. By the Heine-Borel Theorem, each set $B(p, w)$ is compact. By, Weierstrass's Theorem, each continuous function on a compact set has a maximizer. $\square$

Proposition 18. Let $\succsim$ be a rational, locally non-satiated, and continuous preference and $(\mathcal{B}_{++}^W, C^\succsim)$ its associated Walrasian choice structure. Then,

1. $C^\succsim$ is homogeneous of degree zero, i.e., for each $(p, w) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$ and each $\alpha \in \mathbb{R}_{++}$, $C^\succsim(\alpha p, \alpha w) = C^\succsim(p, w)$.
2. $(\mathcal{B}_{++}^W, C^\succsim)$ satisfies Walras' law.
3. If $\succsim$ is convex then $C^\succsim$ is convex-valued, i.e., for each $(p, w) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$, $C^\succsim(p, w)$ is convex. If $\succsim$ is strictly convex then $C^\succsim$ is single-valued.

Proof. Problem 1, Section 3.5.7. $\square$

3.5.4 Continuity of choice

The following result is a direct consequence of a mathematical result known as the Theorem of the Maximum (or Maximum Theorem).

Proposition 19. Let $\succsim$ be a continuous preference on $X \equiv \mathbb{R}_+^L$. Assume that $C^\succsim$ is single-valued. Then, the demand function associated with $(\mathcal{B}_{++}^W, C^\succsim)$ is continuous.

Mathematical aside: The Theorem of the Maximum

For details and proofs of this section see [Sundaram \(1996\)](#).

Let $S \subseteq \mathbb{R}^L$ and $\Theta \subseteq \mathbb{R}^k$ for some $k \in \mathbb{N}$.

Definition 76. A correspondence $\Gamma : \Theta \rightrightarrows S$ is **compact-valued** if for each $\theta \in \Theta$, $\Gamma(\theta)$ is compact.

Definition 77. A non-empty and compact-valued correspondence $\Gamma : \Theta \rightrightarrows S$ is **upper semi-continuous (usc) at $\theta \in \Theta$** if for each convergent sequence in Θ , $\theta_t \xrightarrow[t \rightarrow \infty]{} \theta$ and each sequence in S , $\{s_t\}_{t \in \mathbb{N}}$, such that for each $t \in \mathbb{N}$, $s_t \in \Gamma(\theta_t)$, there is a convergent subsequence $s_{m(t)} \xrightarrow[t \rightarrow \infty]{} s$ such that $s \in \Gamma(\theta)$.

Definition 78. A non-empty and compact-valued correspondence $\Gamma : \Theta \rightrightarrows S$ is **lower semi-continuous (lsc) at $\theta \in \Theta$** if for each convergent sequence in Θ , $\theta_t \xrightarrow[t \rightarrow \infty]{} \theta$, and each $s \in \Gamma(\theta)$, there is a sequence in S , $\{s_t\}_{t \in \mathbb{N}}$, such that for each $t \in \mathbb{N}$, $s_t \in \Gamma(\theta_t)$, and $s_t \xrightarrow[t \rightarrow \infty]{} s$.

Definition 79. A non-empty and compact-valued correspondence $\Gamma : \Theta \rightrightarrows S$ is **continuous at $\theta \in \Theta$** if it is both lsc and usc at θ ; Γ is **continuous** if it is continuous at each point in its domain.

Theorem 13 (Theorem of the Maximum). Let $u : S \times \Theta \rightarrow \mathbb{R}$ be a continuous function, and $\Gamma : \Theta \rightrightarrows S$ a non-empty and compact-valued continuous correspondence. Let V^u and Γ^* be defined by: for each $\theta \in \Theta$,

$$V^u(\theta) \equiv \max_{x \in \Gamma(\theta)} u(x),$$

$$\Gamma^*(\theta) \equiv \arg \max_{x \in \Gamma(\theta)} u(x).$$

Then V^u is continuous and Γ^* is usc.

Corollary 4 (Theorem of the Maximum under convexity). Let $u : S \times \Theta \rightarrow \mathbb{R}$ be a continuous and strictly concave function on S for each θ , and $\Gamma : \Theta \rightrightarrows S$ a non-empty convex- and compact-valued continuous correspondence. Let Γ^* be defined by: for each $\theta \in \Theta$,

$$\Gamma^*(\theta) \equiv \arg \max_{x \in \Gamma(\theta)} u(x).$$

Then Γ^* is a continuous function.

3.5.5 Constrained maximization

The theorem of Kuhn-Tucker provides conditions that allow us to identify points of maxima. You can find a complete treatment of the topic and a more general version of the theorem in [Sundaram \(1996\)](#).

Theorem 14 (Kuhn-Tucker). Let $u : \mathbb{R}^L \rightarrow \mathbb{R}$ and $h_i : \mathbb{R}^L \rightarrow \mathbb{R}$, $i = 1, \dots, I$, be continuously differentiable functions. Suppose that x^* is a local maximizer of u on the set $\{x \in \mathbb{R}^L : h_i(x) \leq 0, i = 1, \dots, I\}$. Let $E \subseteq \{1, \dots, I\}$ be the set of indices such that $h_i(x^*) = 0$, and $h_E \equiv (h_i)_{i \in E}$. Suppose that $\text{Rank}(Dh_E(x^*)) = |E|$. Then, there exists a vector $\lambda^* \equiv (\lambda_1^*, \dots, \lambda_I^*) \in \mathbb{R}^I$ such that the following conditions (Kuhn-Tucker First Order Conditions) are met:

$$\text{For each } i = 1, \dots, I, \lambda_i^* \geq 0 \text{ and } \lambda_i^* h_i(x^*) = 0. \quad (3.1)$$

$$Du(x^*) = \sum_{i=1}^I \lambda_i^* Dh_i(x^*). \quad (3.2)$$

Notice that the theorem, in this basic form, is far from providing sufficient conditions for the constrained maximization of a function. Further convexity requirements help in this direction.

Theorem 15 (Kuhn-Tucker Theorem under quasi-convexity). Let $u : \mathbb{R}^L \rightarrow \mathbb{R}$ be quasi-concave continuously differentiable and $h_i : \mathbb{R}^L \rightarrow \mathbb{R}$, $i = 1, \dots, I$, be quasi-convex continuously differentiable functions. Suppose that there is x^* and $\lambda^* \equiv (\lambda_1^*, \dots, \lambda_I^*) \in \mathbb{R}^I$ such that the following conditions (Kuhn-Tucker First Order Conditions) are met:

$$\text{For each } i = 1, \dots, I, \lambda_i^* \geq 0 \text{ and } \lambda_i^* h_i(x^*) = 0. \quad (3.3)$$

$$Du(x^*) = \sum_{i=1}^I \lambda_i^* Dh_i(x^*). \quad (3.4)$$

Then, x^* is a solution to the problem:

$$\begin{aligned} & \max_x u(x) \\ & \text{subject to } x \in \mathbb{R}^L \text{ and } h_i(x) \leq 0 \text{ for each } i = 1, \dots, I. \end{aligned}$$

provided one of the following conditions holds: (1) $Du(x^*) \neq 0$ or (2) u is concave.

3.5.6 Indirect utility

Definition 80. Let $\succsim$ be a preference relation represented by a utility function u . The **indirect utility function associated with u** is the function, denoted V^u , that associates with each price vector $p \gg 0$ and each wealth level w the maximum value of utility, with respect to welfare index u , attained at choice problem $B(p, w)$. That is

$$\begin{aligned} V^u : \mathbb{R}_{++}^L \times \mathbb{R}_+ & \rightarrow \mathbb{R} \\ (p, w) & \mapsto V^u(p, w) \equiv \max_{x \in B(p, w)} u(x) \end{aligned}$$

The following proposition states properties of indirect utility functions (continuity follows from the Theorem of the Maximum).

Proposition 20. Let $\succsim$ be a locally non-satiated preference relation on $\mathbb{R}_+^L$ that is represented by a continuous utility function u . Then, V^u is

1. Homogeneous of degree zero.
2. Strictly increasing in wealth and non-increasing in each price component.
3. Quasi-convex, i.e., for each $\bar{u} \in \mathbb{R}$, the set $\{(p, w) \in \mathbb{R}_{++}^L \times \mathbb{R}_+ : V^u(p, w) \leq \bar{u}\}$ is convex.

4. Continuous.

Lemma 3. Let $\succsim$ be a locally non-satiated preference relation on $\mathbb{R}_+^L$ represented by a continuously differentiable utility function u . Assume that $C^\succsim$ is single-valued and let x be the associated demand function. Suppose that for some $p \gg 0$ and $w > 0$, $x(p, w) \gg 0$, that the K-T F.O.C. for the associated UMP are satisfied at $x(p, w)$, and that x is differentiable at (p, w) . Then, $\frac{\partial V^u}{\partial w}(p, w) = \lambda$, where λ is the multiplier associated with the budget constraint $p \cdot x - w \leq 0$.

Proof. Recall that $V^u(p, w) = u(x(p, w))$. Using the chain rule,

$$\frac{\partial V^u}{\partial w}(p, w) = \frac{\partial u}{\partial x_1}(x(p, w)) \frac{\partial x_1}{\partial w}(x(p, w)) + \cdots + \frac{\partial u}{\partial x_L}(x(p, w)) \frac{\partial x_L}{\partial w}(x(p, w)).$$

Thus,

$$\frac{\partial V^u}{\partial w}(p, w) = Du^T(x(p, w)) D_w x(p, w).$$

From K-T FOC $\lambda p = Du(x(p, w))$. Thus,

$$\frac{\partial V^u}{\partial w}(p, w) = \lambda p^T D_w x(p, w).$$

By Walras' law, $p \cdot x(p, w) = w$. Differentiating with respect to w , we get $p^T D_w x(p, w) = 1$. Thus,

$$\frac{\partial V^u}{\partial w}(p, w) = \lambda.$$

□

3.5.7 Problems

1. Prove Proposition 18.
2. Let $\succsim$ be the preference on $\mathbb{R}_+^2$ represented by the utility function

$$u(x_1, x_2) \equiv \min\{2x_1, x_2\}.$$

- (a) Draw the indifference curve that passes through $(1, 2)$.
- (b) Calculate the demand function associated with this preference, i.e., for each $p \equiv (p_1, p_2) \in \mathbb{R}_{++}^2$ and each $w \in \mathbb{R}_+$, calculate $x_1(p, w)$ and $x_2(p, w)$.
- (c) Calculate $S_{12}(p, w)$ and $S_{21}(p, w)$. Interpret the result.

- (d) State the compensated law of demand. Prove that x satisfies the compensated law of demand.
3. Let $\succsim$ be the preference relation on $\mathbb{R}_+^2$ represented by the following utility function: for each $(x_1, x_2) \in \mathbb{R}_+^2$,

$$u(x_1, x_2) \equiv \min\{x_1, x_2\}.$$

- (a) Draw three different indifference curves.
- (b) Let $x^\succsim$ be the demand function associated with $\succsim$. Provide a formal definition of $x^\succsim$. Let $p \in \mathbb{R}_{++}^2$ and $w \in \mathbb{R}_+$. Calculate $x^\succsim(p, w) \equiv (x_1^\succsim(p, w), x_2^\succsim(p, w))$.
- (c) What is the value of the Slutsky matrix associated with $x^\succsim$ at some arbitrary $p \in \mathbb{R}_{++}^2$ and $w \in \mathbb{R}_+$.
4. Let $\succsim$ be the preference relation on $\mathbb{R}_+^2$ represented by the following utility function: for each $(x_1, x_2) \in \mathbb{R}_+^2$,

$$u(x_1, x_2) \equiv x_1^\alpha x_2^{1-\alpha},$$

where $\alpha \in (0, 1)$.

- (a) Draw three different indifference curves.
- (b) Let $x^\succsim$ be the demand function associated with $\succsim$. Provide a formal definition of $x^\succsim$. Let $p \in \mathbb{R}_{++}^2$ and $w \in \mathbb{R}_+$. Calculate $x^\succsim(p, w) \equiv (x_1^\succsim(p, w), x_2^\succsim(p, w))$.
- (c) What is the value of the Slutsky matrix associated with $x^\succsim$ at some arbitrary $p \in \mathbb{R}_{++}^2$ and $w \in \mathbb{R}_+$.
5. Let $\succsim_{lex}$ be the lexicographic relation on $\mathbb{R}_+^2$ defined in Section 3.3.
- (a) Draw three different indifference sets.
- (b) Let x^{lex} be the demand function associated with $\succsim_{lex}$. Provide a formal definition of x^{lex} . Let $p \in \mathbb{R}_{++}^2$ and $w \in \mathbb{R}_+$. Calculate $x^{lex}(p, w) \equiv (x_1^{lex}(p, w), x_2^{lex}(p, w))$.
- (c) What is the value of the Slutsky matrix associated with $x^\succsim$ at some arbitrary $p \in \mathbb{R}_{++}^2$ and $w \in \mathbb{R}_+$.
- (d) Is there a continuous utility function u whose associated preference relation $\succsim^u$ is such that $x^{\succsim^u} = x^{lex}$?

6. Consider the problem of maximizing the function $x \equiv (x_1, x_2) \in \mathbb{R}^2 \mapsto u(x) \equiv -\left(x_1 - \frac{9}{4}\right)^2 - (x_2 - 2)^2$ subject to: (1) $x_2 - x_1^2 \geq 0$, (2) $x_1 + x_2 \leq 6$, (3) $x \in \mathbb{R}_+^2$, i.e., $x_1 \geq 0$ and $x_2 \geq 0$.
- (a) Write the Kunh-Tucker F.O.C for this problem.
 - (b) Verify that at $x^* \equiv \left(\frac{3}{2}, \frac{9}{4}\right)$ the K-T F.O.C are satisfied.
 - (c) Prove that x^* is the unique constrained maximizer.
7. Let $\succsim$ be a monotone and continuous preference on $\mathbb{R} \times \mathbb{R}_+^{L-1}$ that is quasi-linear with respect to commodity k and whose associated demand correspondence is single-valued. Let x be the associated demand function.

Let $l \neq k$. The **Marshallian demand for good l** is the function, denoted x_l^M , defined by:

$$\begin{array}{ccc} x_l^M : & \mathbb{R}_{++}^L & \rightarrow \mathbb{R}_+^L \\ & p & \mapsto x_l^M(p) \equiv x(p, w) \text{ for some } w \in \mathbb{R}_{++}. \end{array}$$

Prove that the Marshallian demand for good l is well defined, i.e., for each $p \gg 0$ and each $\{w, w'\} \subseteq \mathbb{R}_{++}$, $x_l(p, w) = x_l(p, w')$.

3.5.8 Expenditure minimization problem

Definition 81. Let $\succsim$ be a preference relation on $\mathbb{R}_+^L$. The **Expenditure Minimization Problem (EMP)** associated with p and $\succsim$ at x^* is:

$$\begin{aligned} \min \quad & p \cdot x \\ \text{s.t.} \quad & x \geq 0 \text{ and } x \succsim x^*. \end{aligned}$$

Definition 82. Let A be a set and $f : A \rightarrow \mathbb{R}$ a real valued function defined on A . An element $x \in A$ is a **minimizer of f on A** if for each $y \in A$, $f(x) \leq f(y)$. The set of minimizers of f on A is denoted $\arg \min_{x \in A} f(x)$ or alternatively $\arg \min\{f(x) : x \in A\}$.

Definition 83. Let $\succsim$ be a locally non-satiated preference relation on $\mathbb{R}_+^L$ represented by a utility function u . The **Hicksian (compensated) demand correspondence**, denoted h , is defined by:

$$\begin{aligned} h : \mathbb{R}_{++}^L \times \mathbb{R}_+^L &\rightarrow \mathbb{R} \\ (p, x^*) &\mapsto h(p, x^*) \equiv \arg \min_{x \geq 0, x \succsim x^*} p \cdot x \end{aligned}$$

Proposition 21. Suppose that $\succsim$ is a locally non-satiated and continuous preference relation defined on $X \equiv \mathbb{R}_+^L$. Let $p \gg 0$. Then, for each $w \geq 0$,

$$x^* \in C^\succsim(p, w) \Leftrightarrow [p \cdot x^* = w \text{ and } x^* \in h(p, x^*)].$$

Definition 84. Let $\succsim$ be a preference relation on $\mathbb{R}_+^L$. The **expenditure function**, denoted e , is defined by:

$$\begin{aligned} e : \mathbb{R}_{++}^L \times \mathbb{R}_+^L &\rightarrow \mathbb{R} \\ (p, x^*) &\mapsto e(p, x^*) \equiv \min_{x \geq 0, x \succsim x^*} p \cdot x \end{aligned}$$

Corollary 5. let $\succsim$ be a locally non-satiated and continuous preference relation on $X \equiv \mathbb{R}_+^L$ that induces a single valued choice correspondence x . Then, h is single valued and for each $p \gg 0$, each $w \geq 0$, and each $x^* \in \mathbb{R}_+^L$,

1. $e(p, x^*) = p \cdot h(p, x^*)$
2. $e(p, x(p, w)) = w$
3. $h(p, x^*) = x(p, e(p, x^*))$
4. $x(p, w) = h(p, x(p, w))$

Proposition 22. Let $\succsim$ be a locally non-satiated and continuous preference relation on $\mathbb{R}_+^L$. Then, e is:

1. Homogeneous of degree one in p , i.e., for each $x^* \in \mathbb{R}_+^L$ and each $\alpha \in \mathbb{R}_{++}$, $e(\alpha p, x^*) = \alpha e(p, x^*)$.
2. Non-decreasing in each price component
3. If $x \succsim x'$, then $e(p, x) \geq e(p, x')$; if $x \succ x'$, then $e(p, x) > e(p, x')$;
4. Concave in p .
5. Continuous in p and x^* .

Proof. We prove that e is concave in p . Let $x^* \in \mathbb{R}_+^L$, $p \gg 0$, $p' \gg 0$, and $\alpha \in [0, 1]$. We want to prove that

$$e(\alpha p + (1 - \alpha)p', x^*) \geq \alpha e(p, x^*) + (1 - \alpha)e(p', x^*). \quad (3.5)$$

By definition, $e(\alpha p + (1 - \alpha)p', x^*) = (\alpha p + (1 - \alpha)p') \cdot y$ where $y \in h(\alpha p + (1 - \alpha)p', x^*)$. Thus, $y \geq 0$ and $y \succsim x^*$. Thus, $e(p, x^*) \leq p \cdot y$ and $e(p', x^*) \leq p' \cdot y$. This implies (3.5). $\square$

Proposition 23. Let $\succsim$ be a preference relation on $\mathbb{R}_+^L$. Then, h is:

1. Homogeneous of degree zero in p , i.e., for each $x^* \in \mathbb{R}_+^L$ and each $\alpha \in \mathbb{R}_{++}$, $h(\alpha p, x^*) = h(p, x^*)$.
2. Let $x^* \in \mathbb{R}_+^L$. If $h(\cdot, x^*)$ is differentiable, then for each $p \gg 0$, $D_p h(p, x^*)p = 0$.
3. If $\succsim$ is convex, then h is convex-valued; if $\succsim$ is strictly convex, then h is single-valued.

Proposition 24. Let $\succsim$ be a locally non-satiated and continuous preference relation on $\mathbb{R}_+^L$. If h is single valued, then it satisfies the compensated law of demand. That is, for any two price vectors, $p \gg 0$ and $p' \gg 0$, and each $x^* \in \mathbb{R}_+^L$,

$$(p' - p) \cdot (h(p', x^*) - h(p, x^*)) \leq 0,$$

with strict inequality whenever $h(p', x^*) \neq h(p, x^*)$.

Proposition 25. Let $\succsim$ be a locally non-satiated and continuous preference relation on $\mathbb{R}_+^L$. Let $x^* \in \mathbb{R}_+^L$. Assume that $h(\cdot, x^*)$ is single-valued. Then, e is differentiable with respect to prices and for each $p \gg 0$ and each $l \in \{1, \dots, L\}$,

$$\frac{\partial e}{\partial p_l}(p, x^*) = h_l(p, x^*),$$

or in matrix notation $D_p e(p, x^*) = h(p, x^*)^T$.³

Intuition. If e is differentiable, then the function $p' \in \mathbb{R}_{++}^L \mapsto \varphi(p') \equiv p' \cdot h(p, x^*) - e(p', x^*)$ is differentiable and has a minimum at $p' = p$. Thus, $D_p \varphi(p) = h(p, x^*)^T - D_p e(p, x^*) = 0$. $\square$

Proposition 26. Let $\succsim$ be a locally non-satiated and continuous preference relation on $\mathbb{R}_+^L$. Let $x^* \in \mathbb{R}_+^L$. Assume that for each $x^* \in \mathbb{R}_+^L$, $h(\cdot, x^*)$ is single-valued and C^1 .⁴ Then, for each $p \gg 0$,

1. $D_p h(p, x^*) = D_p^2 e(p, x^*)$.
2. $D_p h(p, x^*)$ is symmetric.
3. $D_p h(p, x^*)$ is negative semi-definite.
4. $D_p h(p, x^*) \cdot p = 0$.

Proposition 27. Let $\succsim$ be a locally non-satiated and continuous preference relation on $\mathbb{R}_+^L$. Suppose that the associated choice correspondence is single-valued with demand function x . Assume that x is C^1 . Then, for each $p \gg 0$, $w > 0$, and each pair $\{l, k\} \subseteq \{1, \dots, L\}$,

$$S_{lk}(p, w) = \frac{\partial h_l}{\partial p_k}(p, x^*)$$

where $x^* \equiv x(p, w)$.

Proof. Let $x^* \equiv x(p, w)$. Recall that $h_l(p, x^*) = x_l(p, e(p, x^*))$. Thus,

$$\frac{\partial h_l}{\partial p_k}(p, x^*) = \frac{\partial x_l}{\partial p_k}(p, e(p, x^*)) + \frac{\partial x_l}{\partial w}(p, e(p, x^*)) \frac{\partial e}{\partial p_k}(p, x^*).$$

By Proposition 25,

$$\frac{\partial e}{\partial p_k}(p, x^*) = h_k(p, x^*).$$

³In alternative gradient notation this equality is $\nabla_p e(p, x^*) = h(p, x^*)$.

⁴A function is C^1 if its first derivatives exist and are continuous functions. Since h is C^1 , then e is C^2 . This is sufficient for the cross derivatives of e to be symmetrical.

Finally, since $e(p, x^*) = w$ and $h(p, x^*) = x(p, w)$, then

$$\frac{\partial h_l}{\partial p_k}(p, x^*) = \frac{\partial x_l}{\partial p_k}(p, w) + \frac{\partial x_l}{\partial w}(p, w)x_k(p, w).$$

□

Corollary 6. Let $\succsim$ be a locally non-satiated and continuous preference relation on $\mathbb{R}_+^L$. Suppose that the associated choice correspondence is single-valued with demand function x . Assume that x is C^1 . Then for each $(p, w) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$, $S(p, w)$ is symmetric.

Proposition 28. Let A be an $L \times L$ symmetric matrix and suppose that there is $p \gg 0$ such that $Ap = 0$. Then A is negative semi-definite if and only if the matrix obtained by deleting the l -th row and the l -th column from A for some $l \in \{1, \dots, L\}$, is negative semi-definite.

Proof. Parallel to Theorem M.D.4 ([Mas-Colell et al., 1995](#)). □

Corollary 7. Let $\succsim$ be a locally non-satiated and continuous preference relation on $\mathbb{R}_+^L$. Suppose that the associated choice correspondence is single-valued and its associated demand function is C^1 . Then, for each pair $(p, w) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$,

1. $S(p, w)$ is symmetric.
2. $S(p, w)$ is negative semi-definite.
3. $S(p, w) \cdot p = 0$.
4. The matrix obtained by deleting the l -th row and the l -th column from $S(p, w)$ for some $l \in \{1, \dots, L\}$, is negative semi-definite.

Theorem 16 (Integrability Theorem). Let $(\mathcal{B}^W, C)$ be a single-valued Walrasian choice structure satisfying Walras' law and whose associated demand function x is differentiable. If the partial derivatives of x satisfy some *technical additional conditions*, then symmetry and negative semi-definiteness of the Slutsky matrix of x at each pair of prices and wealth is sufficient for the existence of a utility function u such that $C = C^u$, where C^u is the choice correspondence associated with the preference represented by u .

Proposition 29 (Roy's Identity). Let $\succsim$ be a locally non-satiated preference relation on $\mathbb{R}_+^L$ represented by a continuously differentiable utility function u .

Assume that the indirect utility function V^u is differentiable at $(p, w) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$. Then, for each $l \in \{1, \dots, L\}$,

$$x_l(p, w) = -\frac{\frac{\partial V^u}{\partial p_l}(p, w)}{\frac{\partial V^u}{\partial w}(p, w)}.$$

Proof. Let $x^* \equiv x(p, w)$ and $\bar{u} \equiv u(x^*)$. Observe that for each $p' \gg 0$, $\bar{u} = V^u(p', e(p', x^*))$. Applying the chain rule and differentiating with respect to price of commodity $l \in \{1, \dots, L\}$,

$$0 = \frac{\partial V^u}{\partial p_l}(p, e(p, x^*)) + \frac{\partial V^u}{\partial w}(p, e(p, x^*)) \frac{\partial e}{\partial p_l}(p, x^*).$$

Since $\frac{\partial e}{\partial p_l}(p, x^*) = h_l(p, x^*) = x_l(p, w)$, then rearranging terms we get

$$x_l(p, w) = -\frac{\frac{\partial V^u}{\partial p_l}(p, w)}{\frac{\partial V^u}{\partial w}(p, w)}.$$

□

3.5.9 Problems

1. Let $X \equiv \mathbb{R}^3$. Suppose that a Walrasian demand function x satisfies Walras' law and is such that:

$$x_1(p, w) \equiv a + b \frac{p_1}{p_3} + c \frac{p_2}{p_3},$$

$$x_2(p, w) \equiv d + e \frac{p_1}{p_3} + f \frac{p_2}{p_3}.$$

- (a) Calculate x_3 .
 - (b) Prove x is homogeneous of degree zero.
 - (c) Calculate the Slutsky matrix at some $p \gg 0$ and $w > 0$.
 - (d) Provide the restrictions on the demand parameters that are implied by utility maximization. (Disregard the fact that our results were developed under the assumption $X \equiv \mathbb{R}_+^L$)
2. Let $\succsim$ be a locally non-satiated continuous preference relation. Let $p \gg 0$. The **money metric representation of $\succsim$ with respect to p** is the function, denoted $m\langle \succsim, p \rangle$, defined by:

$$\begin{aligned} m\langle \succsim, p \rangle : \mathbb{R}_+^L &\rightarrow \mathbb{R} \\ x &\mapsto m\langle \succsim, p \rangle(x) \equiv \min_{y \geq 0, y \succsim x} p \cdot y \end{aligned}$$

Let $\succsim$ be a monotone and continuous preference relation. (Here, it would suffice to assume that for each $x \in \mathbb{R}_+^L$, $x \succsim 0$.) Let $p \gg 0$. Prove that the money metric representation of $\succsim$ with respect p represents $\succsim$.

3. Let $\succsim$ be a monotone and continuous preference whose associated demand correspondence is single-valued. Let x be the associated demand function. The **equivalent variation of $\succsim$ from $p \in \mathbb{R}_{++}^L$ to $p' \in \mathbb{R}_{++}^L$ at wealth level $w \in \mathbb{R}_+$** , denoted $EV^\succsim(p, p', w)$ is defined by:

$$EV^\succsim(p, p', w) \equiv V^{m(\succsim, p)}(p', w) - V^{m(\succsim, p)}(p, w).$$

The **compensating variation of $\succsim$ from $p \in \mathbb{R}_{++}^L$ to $p' \in \mathbb{R}_{++}^L$ at wealth level $w \in \mathbb{R}_+$** , denoted $CV^\succsim(p, p', w)$ is defined by:

$$CV^\succsim(p, p', w) \equiv V^{m(\succsim, p')}(p', w) - V^{m(\succsim, p')}(p, w).$$

Let $p \gg 0$ and $w \in \mathbb{R}_+$. Prove that for each $p' \gg 0$, the following three statements are equivalent:

- (a) $x(p', w) \succsim x(p, w)$.
- (b) $EV^\succsim(p, p', w) \geq 0$.
- (c) $CV^\succsim(p, p', w) \geq 0$.

3.6 Welfare evaluation of price changes

In this section we investigate the ranking of price-wealth combinations by an agent who is a preference maximizer. We assume that we only know the agent's choices in the form of a demand function. In principle we could reconstruct a utility function that generates the agent's demand (this is implicit in the proof of the Integrability Theorem). We follow an alternative approach that turns out to be simpler. We concentrate on the ranking of the particular situations of interest, i.e., the ranking of the agent's choices at two price-wealth combinations, say (p, w) and (p', w') .

For the sake of simplicity in the presentation we first analyze the ranking of price-wealth combinations that differ only in their price vector.

Definition 85. Let $\succsim$ be a locally non-satiated continuous preference. Let $p \gg 0$. The **money metric representation of $\succsim$ with respect to p** is the function, denoted m_p , defined by:

$$\begin{aligned} m_p : \mathbb{R}_+^L &\rightarrow \mathbb{R} \\ x &\mapsto m_p(x) \equiv \min_{y \geq 0, y \succsim x} p \cdot y \end{aligned}$$

Proposition 30. Let $\succsim$ be a monotone and continuous preference relation.⁵ Let $p \gg 0$. Then, the money metric representation of $\succsim$ with respect to p represents $\succsim$.

Definition 86. The **equivalent variation of $\succsim$ from $p \in \mathbb{R}_{++}^L$ to $p' \in \mathbb{R}_{++}^L$ at wealth level $w \in \mathbb{R}_+$** is:

$$EV^\succsim(p, p', w) \equiv m_p(x(p', w)) - m_p(x(p, w)).$$

Definition 87. The **compensating variation of $\succsim$ from $p \in \mathbb{R}_{++}^L$ to $p' \in \mathbb{R}_{++}^L$ at wealth level $w \in \mathbb{R}_+$** is:

$$CV^\succsim(p, p', w) \equiv m_{p'}(x(p', w)) - m_{p'}(x(p, w)).$$

Proposition 31. Let $\succsim$ be a monotone and continuous preference whose associated demand correspondence is single-valued. Let x be the associated demand function. Let $p \gg 0$ and $w \in \mathbb{R}_+$. Then, for each $p' \gg 0$, the following three statements are equivalent:

1. $x(p', w) \succsim x(p, w)$.

⁵Here, and in the rest of this section, it would suffice to assume that $\succsim$ is locally non-satiated, continuous, and such that for each $x \in \mathbb{R}_+^L$, $x \succsim 0$.

$$2. EV^{\succsim}(p, p', w) \geq 0.$$

$$3. CV^{\succsim}(p, p', w) \geq 0.$$

For each $p \in \mathbb{R}_+^L$ and $t \in \mathbb{R}_+$ let (p_{-l}, t) be the vector obtained by replacing p_l in p by t .

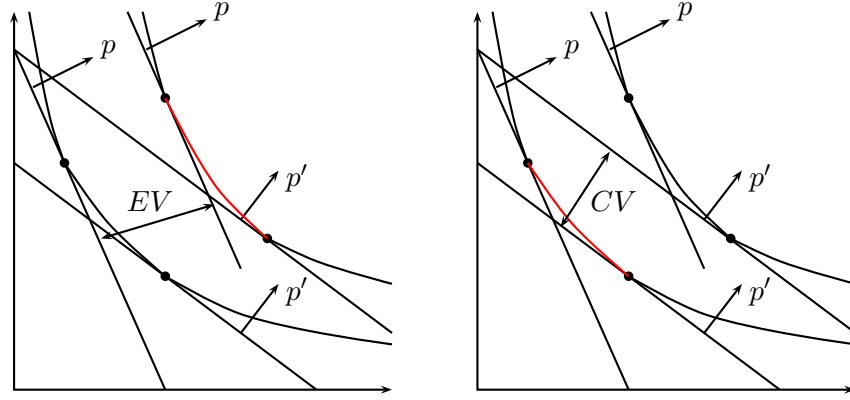


Figure 3.8: Equivalent Variation and Compensating Variation for a change in a single price.

Proposition 32. Let $\succsim$ be a monotone and continuous preference whose associated choice correspondence is single valued. Let $p \gg 0$ and $w \in \mathbb{R}_+$. Then, for each $t \in \mathbb{R}_{++}$

$$EV^{\succsim}(p, (p_{-l}, t), w) = \text{sign}(p_l - t) \int_{\min\{t, p_l\}}^{\max\{t, p_l\}} h_l((p_{-l}, \tau), x((p_{-l}, t), w)) d\tau.$$

Proof.

$$\begin{aligned} EV^{\succsim}(p, (p_{-l}, t), w) &= m_p(x((p_{-l}, t), w)) - m_p(x(p, w)) \\ &= m_p(x((p_{-l}, t), w)) - w \\ &= e(p, x((p_{-l}, t), w)) - e((p_{-l}, t), x((p_{-l}, t), w)) \\ &= e((p_{-l}, p_l), x((p_{-l}, t), w)) - e((p_{-l}, t), x((p_{-l}, t), w)) \end{aligned}$$

Let φ be the function defined by:

$$\begin{aligned} \varphi: \mathbb{R}_+ &\rightarrow \mathbb{R} \\ \tau &\mapsto \varphi(\tau) \equiv e(p_{-l}, \tau, x((p_{-l}, t), w)). \end{aligned}$$

By the Fundamental Theorem of Calculus,

$$\varphi(p_l) - \varphi(t) = \text{sign}(p_l - t) \int_{\min\{t, p_l\}}^{\max\{t, p_l\}} \frac{d\varphi}{d\tau}(\tau) d\tau.$$

Since for each $\tau \in \mathbb{R}_{++}$,

$$\frac{d\varphi}{d\tau}(\tau) = \frac{\partial e}{\partial p_l}((p_{-l}, \tau), x((p_{-l}, t), w)) = h_l((p_{-l}, \tau), x((p_{-l}, t), w)),$$

then

$$EV^{\succsim}(p, (p_{-l}, t), w) = \text{sign}(p_l - t) \int_{\min\{t, p_l\}}^{\max\{t, p_l\}} h_l((p_{-l}, \tau), x((p_{-l}, t), w)) d\tau.$$

□

Corollary 8. Let $\succsim$ be a monotone and continuous preference whose associated choice correspondence is single valued. Let $p \gg 0$ and $w \in \mathbb{R}_+$. Then, for each $p' \gg 0$, $EV^{\succsim}(p, p', w)$ is equal to

$$EV^{\succsim}(p, p', w) = \sum_{l=1}^L \text{sign}(p_l - p'_l) \Phi(l),$$

where

$$\Phi(l) \equiv \int_{\min\{p_l, p'_l\}}^{\max\{p_l, p'_l\}} h_l\left((p_{-\{1, \dots, l\}}, p'_{\{1, \dots, l-1\}}, \tau), x((p_{-\{1, \dots, l\}}, p'_{\{1, \dots, l\}}), w)\right) d\tau.$$

Proof. Problem 1, Section 3.7. □

Proposition 33. Let $\succsim$ be a monotone and continuous preference whose associated choice correspondence is single valued. Let $p \gg 0$ and $w \in \mathbb{R}_+$. Then, for each $t \in \mathbb{R}_{++}$

$$CV^{\succsim}(p, (p_{-l}, t), w) = \int_t^{p_l} h_l((p_{-l}, \tau), x(p, w)) d\tau.$$

Proposition 34. Let $\succsim$ be a monotone and continuous preference on $\mathbb{R} \times \mathbb{R}_+^{L-1}$ that is quasi-linear with respect to commodity k and whose associated demand correspondence is single-valued. Let x be the associated demand function. Let $p \gg 0$ and $w \in \mathbb{R}_+$. Then, for each $t \in \mathbb{R}_{++}$ and each $l \neq k$,

$$EV^{\succsim}(p, (p_{-l}, t), w) = CV^{\succsim}(p, (p_{-l}, t), w) = \int_t^{p_l} x_l((p_{-l}, \tau), w) d\tau.$$

Proof. See Problem 7, Section 3.5.7. □

3.6.1 Welfare evaluation of price-wealth changes in a partial equilibrium environment

Let $\succsim$ be a monotone and continuous preference on $\mathbb{R} \times \mathbb{R}_+^{L-1}$ that is quasi-linear with respect to commodity k and whose associated demand correspondence is single-valued. Let x be the associated demand function.

Definition 88. Let $l \neq k$. The **Marshallian demand for good l** is the function, denoted x_l^M , defined by:

$$\begin{aligned} x_l^M : \mathbb{R}_{++}^L &\rightarrow \mathbb{R}_+^L \\ p &\mapsto x_l^M(p) \equiv x(p, w) \text{ for some } w \in \mathbb{R}_{++}. \end{aligned}$$

Proposition 35. Let $l \neq k$. Then, the Marshallian demand for good l is well defined, i.e., for each $p \gg 0$ and each $\{w, w'\} \subseteq \mathbb{R}_{++}$, $x_l(p, w) = x_l(p, w')$.

Proof. See Problem 7, Section 3.5.7. □

Definition 89. Let $p \gg 0$ and $w \in \mathbb{R}_+$. For each $l \neq k$, the **Marshallian consumer surplus** induced by a change in price of commodity l from p_l to p'_l is

$$MCS_l^{\succsim}(p, p'_l) \equiv \int_{p'_l}^{p_l} x_l^M(p_{-l}, \tau) d\tau.$$

Proposition 36. Let $p \gg 0$ and $l \neq k$. Then, for each $p'_l \in \mathbb{R}_{++}$ and each $w \in \mathbb{R}_+$,

$$x(p, w) \succsim x((p_{-l}, p'_l), w) \text{ if and only if } MCS_l^{\succsim}(p, p'_l) \geq 0.$$

3.7 Problems

1. Prove Corollary 8.
2. Let $\succsim$ be a monotone and continuous preference whose associated choice correspondence is single valued. Let $p \gg 0$ and $w \in \mathbb{R}_+$. Prove that for each $p' \gg 0$,

$$EV^{\succsim}(p, p', w) = \sum_{l=1}^L (p_l - p'_l) \int_0^1 h_l(\alpha p + (1 - \alpha)p', x(p', w)) d\alpha.$$

3. Consider the Cobb-Douglas preference $U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$.

- (a) Calculate the demand function associated with this preference $(x(p, w))$ that maximizes preferences in budget set $B(p, w)$.
- (b) Calculate the Hicksian demand function associated with this preference $(h(p, \tilde{x}))$ that minimizes expenditure at prices p among all bundles that are at least as good as $\tilde{x}$.
- (c) Recall that the equivalent variation from $p \in \mathbb{R}_{++}^2$ to $p' \in \mathbb{R}_{++}^2$ for U at wealth level w is

$$EV(p, p', w) = e(p, x(p', w)) - e(p, x(p, w)),$$

where e is the expenditure function for preference U . Prove that for $p_1, \tilde{p}_1, p_2$ and $w \in \mathbb{R}_+$,

$$EV^{\sim}((p_1, p_2), (\tilde{p}_1, p_2), w) = (p_1 - \tilde{p}_1) \int_0^1 h_1((\theta p_1 + (1 - \theta)\tilde{p}_1, p_2), x(\tilde{p}_1, p_2, w)) d\theta.$$

- (d) Verify that (c) holds for a Cobb-Douglas preference. (here you have to calculate EV from its definition and verify that it is equal to the right side of (c).)

3.8 Aggregate demand

We consider now a multy-agent model.

- There are $L \in \mathbb{N}$ commodities.
- There is a set of agents $N \equiv \{1, \dots, n\}$.
- Agents consume bundles of commodities in $\mathbb{R}_+^L$.
- Agent i 's preferences are denoted by $\succsim^i$.
- We assume that the demand correspondence induced by each $\succsim^i$ is single valued, with demand function x^i .

3.8.1 Aggregate demand function

Definition 90. Let $(x^i)_{i \in N}$ be a profile of demand functions. The **summation of demands** associated with this profile is the function:

$$SD : \quad \mathbb{R}_+^L \times \mathbb{R}_+^N \quad \rightarrow \quad \mathbb{R}_+^L \\ (p, w^1, \dots, w^n) \mapsto SD(p, w^1, \dots, w^n) \equiv \sum_{i \in N} x^i(p, w^i).$$

There are two conditions under which we can talk about an “aggregate demand function” instead of the summation of demands.

Our first condition requires the summation of demands depend only on prices and aggregate wealth.

Definition 91. Let $(x^i)_{i \in N}$ be a profile of demand functions and SD its associated summation of demands. We say that SD is an **aggregate demand function** if for each $p \in \mathbb{R}_{++}^L$ and each two profiles of wealth $w \equiv (w^i)_{i \in N}$ and $\bar{w} \equiv (\bar{w}^i)_{i \in N}$ such that $\sum_{i \in N} w^i = \sum_{i \in N} \bar{w}^i$, $SD(p, w) = SD(p, \bar{w})$. For each $p \in \mathbb{R}_{++}^L$ and each aggregate wealth level $\mu \in \mathbb{R}_+$ we conveniently denote

$$AD(p, \mu) \equiv SD(p, w^1, \dots, w^n),$$

where $(w^i)_{i \in N} \in \mathbb{R}_+^N$ is a profile of wealth such that $\sum_{i \in N} w^i = \mu$.

Proposition 37. The aggregate demand function is well defined whenever:

1. Preferences are quasi-linear.
2. For each $i \in N$, $\succsim^i$ is homothetic, and for each pair $\{i, j\} \subseteq N$, $\succsim^i = \succsim^j$.

Proof. See Problem 1, Section 3.8.4. $\square$

Proposition 38. Let $(x^i)_{i \in N}$ be a profile of differentiable demand functions and SD its associated summation of demands. Then SD is an aggregate demand function if and only if for each $p \in \mathbb{R}_{++}^L$, each wealth vector $w \in \mathbb{R}_+^n$, and each $v \in \mathbb{R}^n$ such that $\sum_{i \in N} v_i = 0$,

$$D_w SD(p, w)v = 0.$$

Proof. ($\Rightarrow$) Suppose that for each pair of vectors w and w' such that $\sum_{i \in N} w^i = \sum_{i \in N} \bar{w}^i$, $SD(p, w) = SD(p, \bar{w})$. Let $v \in \mathbb{R}^n$ be such that $\sum_{i \in N} v_i = 0$. Let $w \gg 0$ be such that $w + v \geq 0$. Then, for each $\alpha \in [0, 1]$, $SD(p, w + \alpha v) = SD(p, w)$. Thus,

$$D_w SD(p, w)v = 0.$$

($\Leftarrow$) Suppose that for each w and each $v \in \mathbb{R}^n$ such that $\sum_{i \in N} v^i = 0$, $D_w SD(p, w)v = 0$. Consider the function $\varphi : [0, 1] \rightarrow \mathbb{R}^L$ defined as follows: for each $\alpha \in [0, 1]$, $\varphi(\alpha) \equiv SD(p, \alpha w + (1 - \alpha)\bar{w})$. Application of the chain rule yields that for each $\alpha \in [0, 1]$, $\varphi'(\alpha) = D_w SD(p, \alpha w + (1 - \alpha)\bar{w})(\alpha w - \alpha \bar{w})$. Let $v \equiv \alpha w - \alpha \bar{w}$. Since $\sum_{i \in N} v^i = 0$, then $\varphi'(\alpha) = 0$. Thus, φ is a constant function. Thus, $\varphi(1) = \varphi(0)$. Thus, $SD(p, w) = SD(p, \bar{w})$. $\square$

Corollary 9. Let $(x^i)_{i \in N}$ be a profile of differentiable demand functions and SD its associated summation of demands. Then SD is an aggregate demand function if and only if for each $p \in \mathbb{R}_{++}^L$, each pair of agents $\{i, j\} \subseteq N$, and each pair of individual wealth levels $\{w^i, w^j\} \in \mathbb{R}_+$,

$$\frac{\partial x^i}{\partial w^i}(p, w^i) = \frac{\partial x^i}{\partial w^j}(p, w^j).$$

Proposition 39. Let $(x^i)_{i \in N}$ be a profile of differentiable demand functions and SD its associated summation of demands. Then SD is an aggregate demand function if and only if there is a profile of utility representations $(u^i)_{i \in N}$ such that the associated indirect utility functions have the form

$$(p, w^i) \in \mathbb{R}_{++}^L \times \mathbb{R}_+ \mapsto V^{u^i}(p, w^i) = a_i(p) + b(p)w^i.$$

Besides the conditions above, we can talk about aggregate demand when we assume that the society distributes wealth in a way that is independent of the agents preferences, that is, depends only on prices and aggregate wealth.

Definition 92. A **wealth distribution rule** is a function that associates a distribution of wealth with each pair $(p, \mu) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$. The generic wealth distribution rule is denoted:

$$\begin{aligned} \omega : \mathbb{R}_+^L \times \mathbb{R}_+ &\rightarrow \mathbb{R}_+^N \\ (p, \mu) &\mapsto \omega(p, \mu) \equiv (\omega^1(p, \mu), \dots, \omega^N(p, \mu)) \\ &\text{such that } \sum_{i \in N} \omega^i(p, \mu) = \mu. \end{aligned}$$

Definition 93. Let $(x^i)_{i \in N}$ be a profile of demand functions and ω a wealth distribution rule. The **aggregate demand associated with $(x^i)_{i \in N}$ and ω** is the function

$$\begin{aligned} AD^\omega : \mathbb{R}_+^L \times \mathbb{R}_+ &\rightarrow \mathbb{R}_+^L \\ (p, \mu) &\mapsto AD^\omega(p, \mu) \equiv \sum_{i \in N} x^i(p, \omega^i(p, \mu)). \end{aligned}$$

Definition 94. A wealth distribution rule is **price-independent** if for any $\{p, p'\} \subseteq \mathbb{R}_{++}^L$ and each $\mu \in \mathbb{R}_+$, $\omega(p, \mu) = \omega(p', \mu)$.

Example 11 (A price-independent wealth distribution rule). Let $(\alpha_i)_{i \in N}$ be such that for each $i \in N$, $\alpha_i \equiv \alpha_i \geq 0$ and $\sum_{i \in N} \alpha_i = 1$. The **proportional wealth distribution rule with weights α** is the function:

$$(p, \mu) \in \mathbb{R}_{++}^L \times \mathbb{R}_+ \mapsto (\alpha_1 \mu, \dots, \alpha_n \mu).$$

3.8.2 Aggregate demand and WARP

The following example shows that properties of individual demand functions are not necessarily transferred into Aggregate demand.

Example 12. Let $N \equiv \{1, 2\}$ and ω the proportional wealth distribution rule with weights $\alpha_1 = \alpha_2 = \frac{1}{2}$. AD^ω may not satisfy WARP.

In the rest of this section AD^ω is the aggregate demand associated with a profile of demands $(x^i)_{i \in N}$ and a wealth distribution rule ω . Recall that AD^ω satisfies WARP if and only if it satisfies the compensated law of demand, i.e., for each $\{p, p'\} \subseteq \mathbb{R}_{++}^L$ and each $\mu \in \mathbb{R}_+$,

$$(p' - p) \cdot [AD^\omega(p', p' \cdot AD^\omega(p, \mu)) - AD^\omega(p, \mu)] \leq 0,$$

with strict inequality when $AD^\omega(p', p' \cdot AD^\omega(p, \mu)) \neq AD^\omega(p, \mu)$.

The example above shows that this inequality may be violated even for proportional wealth distribution rules.

Definition 95. A demand function x satisfies the **uncompensated law of demand** if for each pair $\{p, p'\} \subseteq \mathbb{R}_{++}^L$ and each $\mu \in \mathbb{R}_+$,

$$(p' - p) \cdot [x(p', \mu) - x(p, \mu)] \leq 0,$$

with strict inequality when $x(p', \mu) \neq x(p, \mu)$.

Proposition 40. Suppose that ω is price-independent. If for each $i \in N$, x^i satisfies the uncompensated law of demand, then AD^ω satisfies the uncompensated law of demand.

Proof. Let $\{p, p'\} \subseteq \mathbb{R}_{++}^L$ and $\mu \in \mathbb{R}_+$. Let $w \equiv \omega(p, \mu) = \omega(p', \mu)$; since ω is price-independent, then w is well defined. We prove that

$$(p' - p) \cdot \left(\sum_{i \in N} x^i(p', w^i) - \sum_{i \in N} x^i(p, w^i) \right) \leq 0, \quad (3.6)$$

with strict inequality when $\sum_{i \in N} x^i(p', w^i) \neq \sum_{i \in N} x^i(p, w^i)$. Clearly, we can write the inequality above as:

$$\sum_{i \in N} (p' - p) \cdot (x^i(p', w^i) - x^i(p, w^i)) \leq 0.$$

Let $i \in N$. Since x^i satisfies ULD, then $(p' - p) \cdot (x^i(p', w^i) - x^i(p, w^i)) \leq 0$. Thus, the weak inequality (3.6) holds.

Now, if $\sum_{i \in N} x^i(p', w^i) \neq \sum_{i \in N} x^i(p, w^i)$, then it must be the case that for some $i \in N$, $x^i(p', w^i) \neq x^i(p, w^i)$. Thus, $(p' - p) \cdot (x^i(p', w^i) - x^i(p, w^i)) < 0$. Thus, the strict inequality in (3.6) holds. $\square$

Proposition 41. If a demand function x satisfies Walras' law and the uncompensated law of demand, then it satisfies WARP.

Proof. Let $\{p, p'\} \subseteq \mathbb{R}_{++}^L$ and $w \in \mathbb{R}_+$. We prove that

$$(p' - p) \cdot (x(p', p' \cdot x(p, w)) - x(p, w)) < 0,$$

when $x(p', p' \cdot x(p, w)) \neq x(p, w)$. Since $p' \cdot x(p', p' \cdot x(p, w)) = p' \cdot x(p, w)$, then the inequality above is equivalent to

$$-p \cdot (x(p', p' \cdot x(p, w)) - x(p, w)) < 0.$$

Let $w' \equiv p' \cdot x(p, w)$. Suppose without loss of generality that $w' > 0$, for otherwise the problem is trivial. Let $\alpha = \frac{w}{w'}$ and $\tilde{p} \equiv \alpha p'$. Since x

is homogeneous of degree zero, then $x(p', p' \cdot x(p, w)) = x(\tilde{p}, w)$. Then, $x(\tilde{p}, w) \neq x(p, w)$. Since x satisfies ULD, then

$$(\tilde{p} - p) \cdot (x(\tilde{p}, w) - x(p, w)) < 0.$$

Since $\tilde{p} \cdot x(\tilde{p}, w) = w$ and $\tilde{p} \cdot x(p, w) = \frac{w}{w} p' \cdot x(p, w) = w$, then

$$-p \cdot (x(\tilde{p}, w) - x(p, w)) < 0.$$

Equivalently,

$$-p \cdot (x(p', p' \cdot x(p, w)) - x(p, w)) < 0.$$

□

Corollary 10. Suppose that ω is price-independent and for each $i \in N$, x^i satisfies Walras' law and the uncompensated law of demand. Then AD^ω satisfies WARP.

Proposition 42. Suppose that $\succsim$ is homothetic and its associated demand correspondence is single valued with demand x . Then, x satisfies the ULD.

Proposition 43. If a twice differentiable demand function x satisfies the uncompensated law of demand, then for each $(p, \mu) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$, $D_p x(p, \mu)$ is negative semi-definite.

Proposition 44. Let x be a twice differentiable demand function. If for each pair $(p, \mu) \in \mathbb{R}_{++}^L \times \mathbb{R}_{++}$, $D_p x(p, \mu)$ is negative definite, then the demand function x satisfies the uncompensated law of demand.

Proposition 45. Assume that $X \equiv \mathbb{R}_+^2$ and $\succsim$ is a homothetic preference relation such that for each pair $(p, \mu) \in \mathbb{R}_{++}^L \times \mathbb{R}_{++}$ and each vector $v \in \mathbb{R}^L$ that is not multiple scalar of p , $v^T S(p, \mu) v < 0$. Then, for each pair $(p, \mu) \in \mathbb{R}_{++}^L \times \mathbb{R}_{++}$, $D_p x(p, \mu)$ is negative definite and thus, x satisfies the uncompensated law of demand.

Proposition 46. Let ω be the proportional wealth distribution rule with weights $(\alpha_i)_{i \in N}$. For each $i \in N$ and each pair $(p, \mu) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$, let $S_i(p, \mu)$ be the Slutsky matrix associated with x^i at (p, μ) . Then for each $(p, \mu) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$, the Slutsky matrix associated to AD^ω , denoted S , can be calculated as:

$$S(p, \mu) = \sum_{i \in N} S_i(p, \mu) - \sum_{i \in N} \alpha_i A_i B_i^T.$$

where for each $i \in N$,

$$A_i = [D_{w^i} x^i(p, \alpha_i \mu) - D_\mu AD^\omega(p, \mu)],$$

and

$$B_i \equiv \left[\frac{1}{\alpha_i} x^i(p, \alpha_i \mu) - AD^\omega(p, \mu) \right].$$

3.8.3 Aggregate demand and welfare

Is there any welfare information in aggregate demand?

Definition 96. Let $\succsim \equiv (\succsim_i)_{i \in N}$ be a profile of preferences. A **social welfare pair** for $\succsim$ is a pair (u, W) where $u = (u_i)_{i \in N}$ is a profile of representations for $\succsim$ and W is a function from $\mathbb{R}^N$ to $\mathbb{R}$. The **social welfare order associated with (u, W)** is the complete and transitive binary relation on $\mathbb{R}_+^{L \times N}$ represented by the function

$$\begin{aligned} W : \quad \mathbb{R}_+^{L \times N} &\rightarrow \mathbb{R} \\ (x^1, \dots, x^n) &\mapsto W(u_1(x^1), \dots, u_n(x^n)). \end{aligned}$$

Proposition 47. Assume that $(x^i)_{i \in N}$ are the result of the maximization of preferences $\succsim \equiv (\succsim_i)_{i \in N}$. Let (u, W) be a social welfare pair for $\succsim$. Suppose that for each $(p, w) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$, $\omega(p, w)$ solves the problem

$$\begin{aligned} \max & W(u_1(x^1(p, w^1)), \dots, u_1(x^n(p, w^n))) \\ \text{s.t.} & (w^i)_{i \in N} \in \mathbb{R}_+^N, \sum_{i \in N} w^i = w \end{aligned} \quad (3.7)$$

Then, the aggregate demand function AD^ω is the result of the maximization of the preference represented by:

$$\begin{aligned} U : \mathbb{R}_+^L &\rightarrow \mathbb{R} \\ x &\mapsto U(x) \equiv \max_{\text{s.t. } \{y^1, \dots, y^n\} \subseteq \mathbb{R}_+^L, \sum_{i \in N} y^i \leq x} W(u_1(y^1), \dots, u_1(y^n)) \end{aligned}$$

Moreover, the indirect utility function of (3.7) is the indirect utility function associated with this representation of preferences.

Problem 1. Let $(\alpha_i)_{i \in N}$ be a such that for each $i \in N$, $\alpha_i \geq 0$ and $\sum_{i \in N} \alpha_i = 1$. Suppose that consumers have homothetic preferences and (u, W) is a social welfare pair for these preferences such that the utility functions $(u_i)_{i \in N}$ are non-negative homogeneous of degree one, and

$y \in \mathbb{R}^N \mapsto W(y) \equiv \sum_{i \in N} \alpha_i \ln(y^i)$ for . Let $p \gg 0$ and $w > 0$. Prove that $\bar{w} = (\bar{w}^i)_{i \in N}$ solves

$$\begin{aligned} \max & W(u_1(x^1(p, w^1)), \dots, u_1(x^n(p, w^n))) \\ \text{s.t. } & (w^i)_{i \in N} \in \mathbb{R}_+^N, \sum_{i \in N} w^i = w \end{aligned}$$

if and only if it solves

$$\begin{aligned} \max & \sum_{i \in N} \alpha_i \ln(w^i) \\ \text{s.t. } & (w^i)_{i \in N} \in \mathbb{R}_+^N, \sum_{i \in N} w^i = w. \end{aligned}$$

Use KT FOC to prove that $\bar{w} = (\alpha_i w, \dots, \alpha_n w)$.

What can you conclude about the welfare information contained in the aggregate demand associated with a proportional welfare distribution rule?

3.8.4 Problems

1. Prove Proposition 37.
2. Assume that $X \equiv \mathbb{R}_+^2$ and $\succsim$ is a homothetic, monotone, continuous, and strictly convex preference relation.
 - (a) Prove that the associated demand correspondence is single valued.
 - (b) Prove that the associated demand function satisfies the uncompensated law of demand.
3. Consider the preference on $\mathbb{R} \times \mathbb{R}_+$ represented by the following function: for each $(x_1, x_2) \in \mathbb{R} \times \mathbb{R}_+$

$$u(x_1, x_2) = x_1 + \varphi(x_2),$$

where φ is a continuous real-valued function.

Assume that for each $(p, w) \in \mathbb{R}_{++}^2 \times \mathbb{R}_+$ there is a **unique** bundle that maximizes u in $B(p, w)$. Let x be the demand function associated with the preference represented by u . For the sake of simplicity you can normalize prices so the price of the first commodity is always 1.

- (a) State the definition of quasi-linearity with respect to the first commodity (for preferences on $\mathbb{R} \times \mathbb{R}_+$)
- (b) Prove that the preference represented by u is quasi-linear.

- (c) Let $(p, w) \in \mathbb{R}_{++}^2 \times \mathbb{R}_+$ and $w' \in \mathbb{R}_+$ such that $w' > w$. Prove that $x_2(p, w') = x_2(p, w)$. (Here a contradiction works fine, i.e., suppose that $x_2(p, w') \neq x_2(p, w), \dots$)
 - (d) State the Compensated Law of Demand.
 - (e) Does x satisfy the Compensated Law of Demand? (here I am not asking for a proof; you should be able to give a precise answer, i.e., “yes because...” or “no, because....”)
 - (f) State the Uncompensated Law of Demand.
 - (g) Prove that x may not satisfy the Uncompensated Law of Demand.
4. Construct a strictly convex quasi-linear preference in $\mathbb{R} \times \mathbb{R}_+$ whose associated demand function does not satisfy the ULD.
 5. Assume that $X \equiv \mathbb{R}_+^2$ and $\succsim$ is a Cobb-Douglas preference relation. Provide a direct calculation that shows the associated demand function satisfies the uncompensated law of demand.
 6. Assume that $X \equiv \mathbb{R}_+^2$ and $\succsim$ is a Leontief preference relation. Provide a direct calculation that shows the associated demand function satisfies the uncompensated law of demand.
 7. (Additional) Let ω be the proportional wealth distribution rule with weights $(\alpha_i)_{i \in N}$. Suppose that individual demand functions $(x^i)_{i \in N}$ are the results of the maximization of homothetic preferences. Prove that the Slutsky matrix associated with AD^ω evaluated at $(p, \mu) \in \mathbb{R}_{++}^L \times \mathbb{R}_+$, is symmetric and negative semi-definite.
 8. This problem is intended to explore how “fragile” the Uncompensated Law of Demand is.

Let $\succsim$ be the preference on $\mathbb{R}_+^2$ represented by the utility function

$$u(x_1, x_2) \equiv \min\{x_1, x_2\}.$$

Let $x^\succsim$ be the demand function associated with $\succsim$.

- (a) State the Uncompensated Law of Demand
- (b) Prove that $x^\succsim$ satisfies the uncompensated law of demand.
- (c) Now consider the preference on $\mathbb{R}_+^2$, denoted $\succsim^m$, represented by the utility function

$$u^m(x_1, x_2) \equiv \begin{cases} \min\{x_1, x_2\} & \text{if } \min\{x_1, x_2\} \leq 1 \\ \min\left\{x_1, \frac{1}{m}x_2 + \frac{m-1}{m}\right\} & \text{if } \min\{x_1, x_2\} > 1. \end{cases}$$

where $m \in (0, +\infty)$. The following figure illustrates the indifference curves of $\succsim^m$. Prove that for each $m \neq 1$, the demand function associated with $\succsim^m$ violates the Uncompensated law of demand.

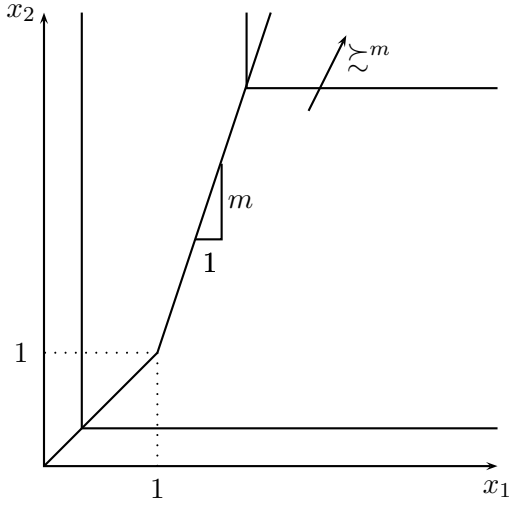


Figure 3.9: Illustrating preferences $\succsim^m$.